

Build for Bharat 2.0 Hackathon:

Approach Notice Submission

Team: Command Q

Team Leader: Rajat Kumar Gupta (25BDA70384)

Career Navigation System for Data Talent Workforce: A Data-Driven Approach to Skills Alignment and Career Progression

Command Q

I. PROBLEM DEFINITION AND ANALYTICS OBJECTIVE

A. Problem Identification

The data science and analytics workforce in India and globally faces a critical structural challenge: aspiring and practicing professionals struggle to navigate career progression effectively due to profound information asymmetry. Despite surging demand for data talent, individuals lack empirical clarity regarding:

- **Personality-Role Congruence:** How foundational psychometric traits dictate long-term success, role sustainability, and progression in high-ambiguity senior technical leadership.
- **Skills Valuation Disconnect:** Which specific competencies drive internal promotion velocity versus advertised hiring demand, avoiding market signaling distortions.
- **Strategic Human Capital Investment:** How to allocate finite reskilling hours to maximize competitive differentiation and wage appreciation.
- **Employer-Experience Alignment:** Which organizations offer optimal compensation brackets and liquidity for an individual's specific career tenure.

This pervasive information asymmetry precipitates career churn, misdirected upskilling expenditure, and wage suppression. Our study provides a rigorous, empirical foundation for an intelligent career navigation system tailored to three primary career cohorts: Senior Data Scientists (SDS), Junior Data Scientists (JDS), and Analytics/BI Professionals.

B. Scope and Analytical Depth

Our analytical architecture integrates four distinct empirical dimensions:

- 1) **Psychometric-Career Alignment:** Quantifying Big Five personality trait effect sizes ($d > 1.70$) and predictive classification bounds for senior data science leadership.
- 2) **Skills Market Intelligence:** Evaluating skill-outcome effect sizes, market penetration rates across 14,749 postings, and econometric wage premiums.

- 3) **Strategic Market Positioning:** Establishing a 2×2 Strategic Skills Matrix and Rank Gap metric to identify undervalued high-leverage skills.
- 4) **Econometric Wage Modeling & Matching:** Developing cross-validated salary regression models ($R^2 = 0.571$) and a multi-attribute organizational matching engine.

C. Analytics Objective and Success Criteria

The primary objective is to build an inspectable, data-driven career navigation engine that operationalizes these findings into actionable career prescriptions. **Measurable Evaluation Criteria:**

- Statistical significance and large effect sizes ($|d| \geq 0.80$, $AUC > 0.75$, $p < 0.001$) across validated traits and skills.
- Cross-validated classification accuracy ($> 85\%$ ROC-AUC) demonstrating high discriminative validity.
- High goodness-of-fit ($R^2 > 0.50$, MAE reduction $> 35\%$) in baseline salary modeling.
- Total reproducibility where every metric, decision tree rule, and regression coefficient is traced to open code listings and verified console outputs.

II. APPROACH DESCRIPTION AND PROTOTYPE ARCHITECTURE

A. Overall System Architecture

Our proposed prototype, the **Career Navigation System for Data Talent Workforce**, translates empirical findings into a four-stage interactive career navigation pipeline, visualized in Fig. 1.

The architecture couples user assessment inputs with four specialized analytical engines:

- 1) **Career Track Norming & Psychometric Fit Engine:** Standardizes candidate profile inputs against benchmark cohorts.
- 2) **Strategic Skills Recommendation Engine:** Evaluates proficiency against the 2×2 Strategic Market Matrix.

Career Navigation System: Analytical Engine Architecture

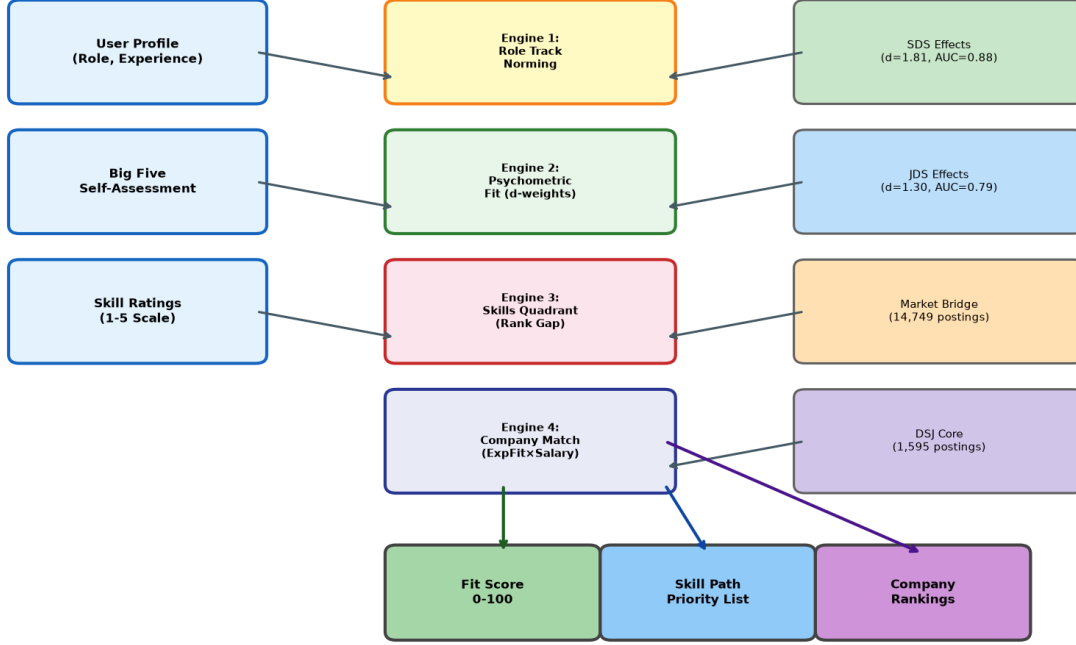


Fig. 1. Career Navigation System: End-to-End Modular Prototype Architecture.

- 3) **Progression Path Optimizer:** Traverses empirical skill co-development graphs to design personalized learning sequences.
- 4) **Multi-Attribute Company Matching Engine:** Matches candidates to hiring organizations optimizing experience fit, salary bracket, and vacancy liquidity.

B. Mathematical Formulations of Recommendation Engines

1) *Engine 1: Psychometric Fit Engine:* To assess alignment with senior data science leadership without arbitrary heuristics, we formulate a normalized psychometric fit index ($S_{\text{fit}} \in [0, 100]$):

$$S_{\text{fit}} = 100 \times \sum_{i=1}^5 w_i \cdot \max\left(0, \min\left(1, \frac{x_i - \mu_{\text{low},i}}{\mu_{\text{high},i} - \mu_{\text{low},i}}\right)\right) \quad (1)$$

where trait weights w_i are proportional to empirical effect sizes (Cohen's d_i):

$$w_i = \frac{|d_i|}{\sum_{j=1}^5 |d_j|} \quad (2)$$

Empirically calibrated weights from our benchmark cohort ($N = 161$) yield:

- Conscientiousness: $w_C = 0.30$
($\mu_{\text{high}} = 53.68, \mu_{\text{low}} = 35.74$)

- Openness to Experience: $w_O = 0.29$
($\mu_{\text{high}} = 48.49, \mu_{\text{low}} = 33.32$)
- Extraversion: $w_E = 0.19$
($\mu_{\text{high}} = 48.86, \mu_{\text{low}} = 36.88$)
- Agreeableness: $w_A = 0.10$
($\mu_{\text{high}} = 47.72, \mu_{\text{low}} = 41.12$)
- Neuroticism: $w_N = 0.12$ (Penalizes extreme scores > 50)

2) *Engine 2: Strategic Skills Recommendation Engine:* To balance internal promotion return against external market employability, the system computes a Skill Priority Score P_k for candidate skill deficits:

$$P_k = \alpha \cdot d_k + \beta \cdot (1 - D_k) + \gamma \cdot \text{RankGap}_k \quad (3)$$

where d_k is the standardized outcome effect size, $D_k \in [0, 1]$ is the macro market demand rate, and $\text{RankGap}_k = \text{DemandRank}_k - \text{EffectRank}_k$. Calibrated hyperparameters ($\alpha = 0.45, \beta = 0.30, \gamma = 0.25$) prioritize “Hidden Gem” competencies that unlock disproportionate promotion velocity.

3) *Engine 3: Progression Path Optimizer:* Reskilling paths are optimized via a directed acyclic graph $G = (V, E)$, where vertices represent skills and edge weights represent empirical

co-development correlations r_{ij} :

$$W(e_{ij}) = r_{ij} \cdot \left(1 + \mathbb{I}(\text{Exp} \geq 4) \cdot \Delta D_{\text{senior},j}\right) \quad (4)$$

This accounts for the structural demand shift where Big Data infrastructure skills become mandatory only beyond Year 4.

4) *Engine 4: Multi-Attribute Company Matching Engine:* For candidate experience E , the system scores prospective hiring organizations c :

$$\begin{aligned} M_c &= 0.50 \cdot \text{ExpFit}(E, c) \\ &+ 0.35 \cdot \text{SalaryScore}(c) \\ &+ 0.15 \cdot \text{LiquidityScore}(c) \end{aligned} \quad (5)$$

where, writing $m_c = \text{MinExp}_c$:

$$\begin{aligned} \text{ExpFit}(E, c) &= \begin{cases} 1.0, & \text{if } m_c \leq E \leq m_c + 2, \\ \max(0, 1 - 0.25 |m_c - E|), & \text{otherwise,} \end{cases} \\ \text{SalaryScore}(c) &= \text{Percentile}(\text{AvgSalary}_c \mid \text{Seniority Tier}), \\ \text{LiquidityScore}(c) &= \frac{\ln(1 + \text{Openings}_c)}{\max_j \ln(1 + \text{Openings}_j)}. \end{aligned}$$

C. User Personas and Journey Flows

The prototype supports three archetypal user journeys:

- **Persona 1: Junior Aspiring Data Scientist (1–2 YOE):** Assesses foundational coding/ML skills. Identifies Storytelling deficit (RankGap = +2). Prescribes data narrative modules; routes toward high-hike organizations.
- **Persona 2: Mid-Career Business/Data Analyst (3–5 YOE):** Tests transition potential to Data Science. Flags math/statistics gap and identifies Big Data timing inflection. Targets firms paying +72% senior analyst premiums.
- **Persona 3: Senior Technical Specialist (7+ YOE):** Evaluates transition into Lead Data Scientist or Data Architect roles. Psychometric engine assesses Conscientiousness and Openness, identifying leadership and architectural readiness.

III. DATA EXPLORATION AND PREPROCESSING

A. Data Sources Overview

Our empirical investigation utilizes four core datasets comprising 16,741 raw records:

B. Data Quality Assessment and Treatment Policy

Prior to modeling, we conducted systematic audits across schema completeness, key integrity, duplicate rows, and domain constraints [Listing G.3 & Output G.3]. Crucially, our analysis verified that ID columns across datasets represent coincidental numerical overlaps rather than foreign keys [Listing G.4 & Output G.4]. Analysis therefore proceeds at the construct level.

TABLE I
SUMMARY OF EMPIRICAL DATASETS

Dataset	Core Construct	Raw Rows	Core Rows
SDS Personality	Big Five traits (OCEAN), Success Flag	161	161
JDS Skills	5 Skill Proficiencies, Salary Hike Flag	139	139
Data Science Jobs	Experience, Salary LPA, Openings	1,602	1,595
Analytics Jobs	Designations, Skills, Experience, Salary	14,839	14,749

TABLE II
DATA CLEANING AND TREATMENT POLICY

Source	Raw	Clean	Dropped	Rationale
JDS Skills	139	139	0 (0.0%)	Fully valid scores (1–5)
SDS Personality	161	161	0 (0.0%)	Fully valid traits (17–68)
Data Science Jobs	1,602	1,595	7 (0.4%)	Placeholder company flags
Analytics Jobs	14,839	14,749	90 (0.6%)	Gig/data-entry deduplication

As documented in Table II and verified in [Listing G.5 & Output G.5], we applied conservative treatments that retained > 99.4% of records while purging corrupt records and gig postings.

C. Distributional Characteristics and Normality Testing

We conducted Shapiro-Wilk normality tests on all numerical attributes [Listing G.7 & Output G.7]. Every single trait and skill score rejected normality at $p < 0.001$ (e.g., JDS Coding $W = 0.788, p = 1.2 \times 10^{-13}$; SDS Conscientiousness $W = 0.921, p = 8.9 \times 10^{-8}$). Consequently, we anchored all subsequent hypothesis testing in non-parametric Mann-Whitney U formulations.

The empirical distributions are presented across Figs. 2, 3, 4, 5, and 6.

D. Feature Derivation and Construct Grounding

We engineered standardized analytical constructs [Listing G.6 & Output G.6]:

- **Role Family Categorization:** Clustered unstructured job designations into 6 core tracks: Analyst ($n = 4,007$), Consultant/Manager ($n = 3,726$), Data Engineer ($n = 446$), Data Scientist ($n = 439$), Data Architect ($n = 357$), and AI/ML Engineer ($n = 201$).
- **Skill Dimension Extraction:** Extracted standardized keyword vectors for Big Data, Math/Stats, Coding, AI/ML, and Dashboard/Storytelling.
- **Seniority Standardization:** Mapped experience into Junior (0–3 yrs), Mid (3–7 yrs), and Senior (8+ yrs).

IV. IMPLEMENTATION PIPELINE AND REPRODUCIBILITY

The entire analytical workflow is organized into an inspectable, staged Python architecture [Listing G.1], separating data ingestion [Listing G.2 & Output G.2], auditing, transformation, statistical testing, multivariate modeling, and artifact export [Listing G.13 & Output G.13].

SDS trait distributions

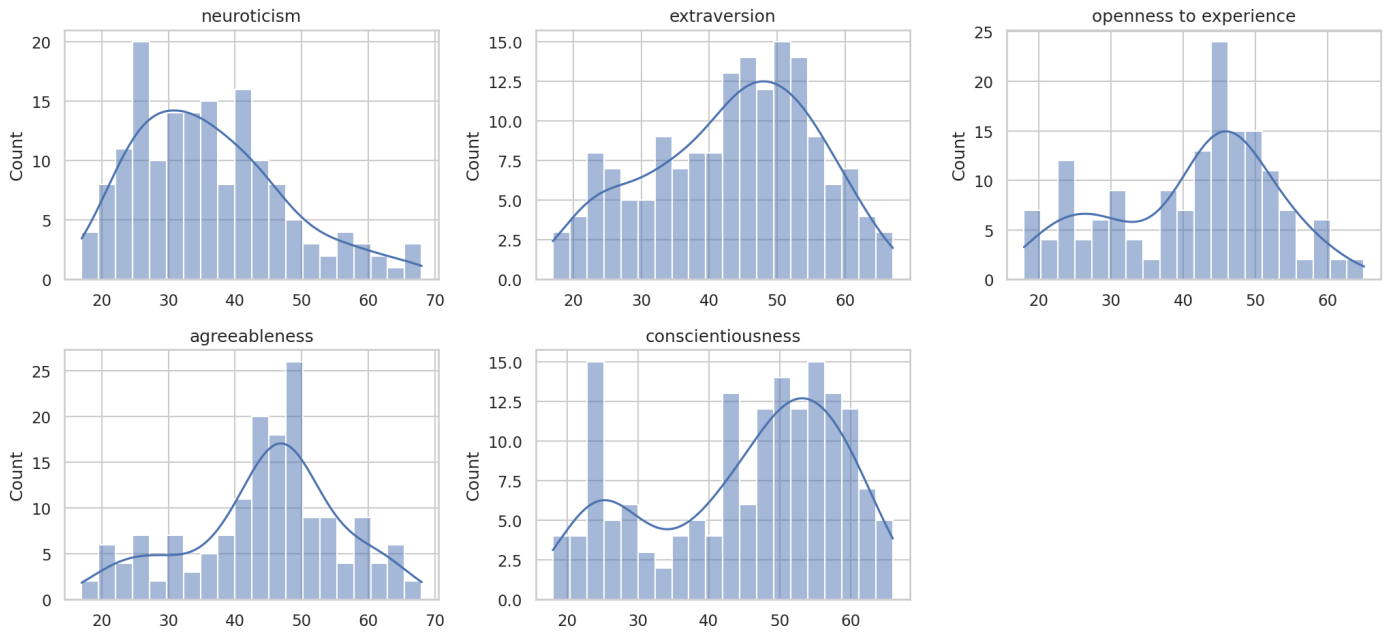


Fig. 2. Distribution of Big Five Personality Traits for Senior Data Scientists.

JDS skill distributions

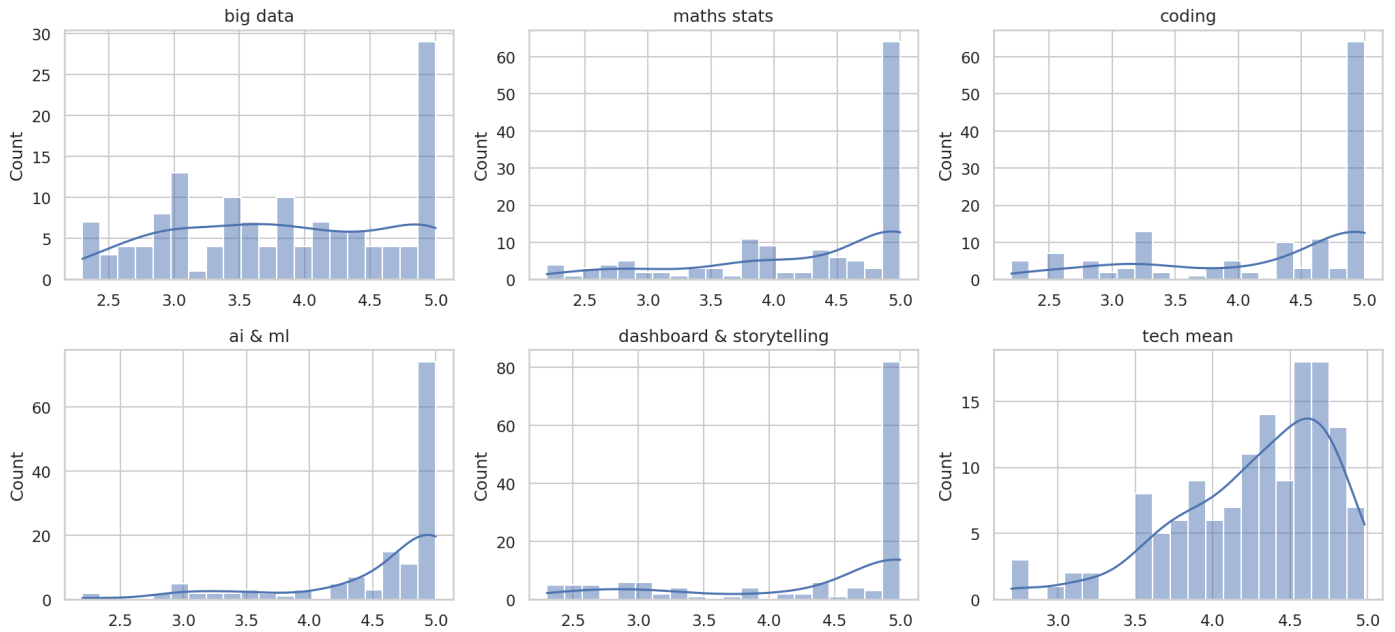


Fig. 3. Proficiency Distributions Across Five Core Junior Data Science Skills.

All derived datasets, bridge tables, and high-resolution figures are automatically generated and validated via the final recap engine [Listing G.14 & Output G.14], ensuring full reproducibility across research and enterprise deployments.

DataScience Jobs distributions

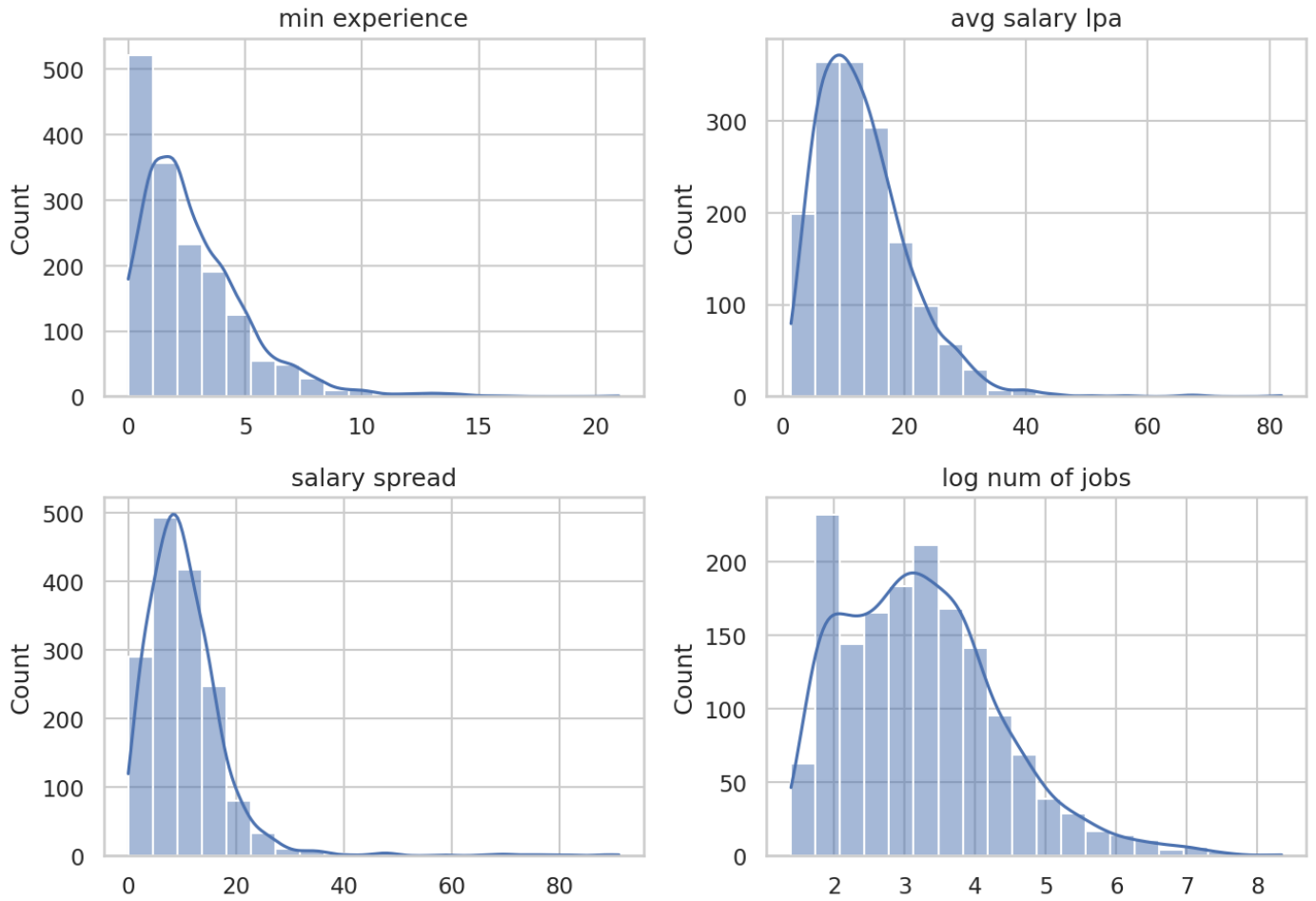


Fig. 4. Data Science Jobs: Salary LPA and Experience Distributions.

Analytics Jobs distributions



Fig. 5. Analytics Jobs: Market Experience and Salary Overview.

V. DATA ANALYSIS AND EMPIRICAL FOUNDATIONS

A. Personality-Career Alignment Analysis

1) *Non-Parametric Group Differences:* Evaluating Big Five personality scores between high-performing senior data scien-

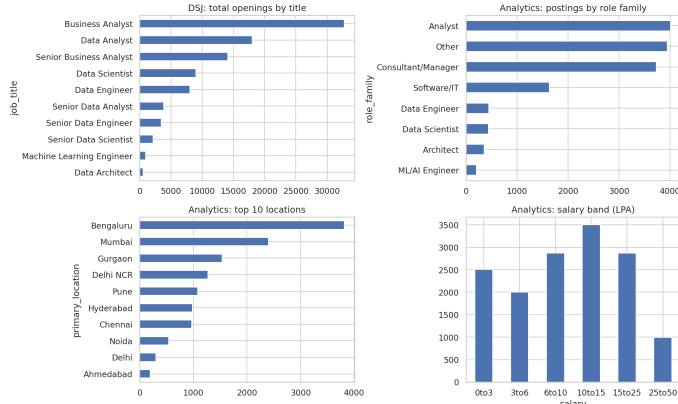


Fig. 6. Categorical Feature Distributions Across Job Market Postings.

TABLE III
PERSONALITY TRAIT EFFECT SIZES FOR SENIOR DATA SCIENTISTS

Trait	High μ	Low μ	Cohen's d	AUC	Bonferroni p
Conscientiousness	53.68	35.74	1.815	0.878	1.44×10^{-16} *
Openness to Exp.	48.49	33.32	1.774	0.885	3.64×10^{-17} *
Extraversion	48.86	36.88	1.115	0.783	5.79×10^{-10} *
Agreeableness	47.72	41.12	0.595	0.652	0.004*
Neuroticism	36.13	36.26	-0.012	0.534	1.000 (ns)

TABLE IV
SDS PREDICTIVE CLASSIFICATION BENCHMARKS (5-FOLD, 10-REPEAT CV)

Model Architecture	CV ROC-AUC	AUC SD	CV Accuracy
Baseline (Majority Class)	0.500	0.000	52.8%
Logistic Regression (Standardized)	0.964	0.033	92.7%
Decision Tree (Depth 3)	0.942	0.039	93.4%
Random Forest ($B = 300$)	0.996	0.008	94.9%

tists ($n = 85$) and the comparison cohort ($n = 76$) reveals massive, statistically robust effect sizes (Table III, verified in [Listing G.8 & Output G.8a]).

Conscientiousness ($d = 1.815$, AUC = 0.878) and Openness to Experience ($d = 1.774$, AUC = 0.885) demonstrate extraordinary discriminatory power. Senior technical leadership requires intellectual curiosity to formulate non-standard algorithmic approaches coupled with rigorous conscientiousness to enforce pipeline reliability and business delivery.

2) *Repeated Cross-Validation Model Benchmarks:* To test predictive validity, we evaluated models under 5-fold cross-validation with 10 repeated resamplings (50 total train/test evaluations) [Listing G.11 & Output G.11a].

Standardized multivariate logistic regression reveals that a 1-SD increase in Conscientiousness multiplies the odds of senior leadership success by **8.11** ($\beta = 2.094$), while Openness multiplies odds by **7.72** ($\beta = 2.043$) (Fig. 9).

3) *Interpretable Decision Heuristics:* Extracting the depth-3 decision tree rules provides transparent heuristics for our career navigation system:

```
SDS Classification Heuristic (Accuracy = 93.4%):
| --- Openness to Experience <= 38.50
| | --- class: 0 (Low Senior Success)
```

TABLE V
SKILL EFFECT SIZES FOR JUNIOR DATA SCIENTISTS

Skill Dimension	High μ	Low μ	Cohen's d	AUC	Bonferroni p
Dashboard & Storytelling	4.845	3.814	1.302	0.794	2.40×10^{-11} *
Math & Statistics	4.712	3.830	1.202	0.775	6.66×10^{-9} *
Coding	4.644	3.853	0.976	0.744	2.03×10^{-7} *
AI & Machine Learning	4.822	4.283	0.863	0.682	8.57×10^{-5} *
Big Data Infrastructure	3.940	3.750	0.223	0.561	0.217 (ns)

TABLE VI
JDS PREDICTIVE CLASSIFICATION BENCHMARKS (5-FOLD, 10-REPEAT CV)

Model Architecture	CV ROC-AUC	AUC SD	CV Accuracy
Baseline (Majority Class)	0.500	0.000	52.5%
Logistic Regression (Standardized)	0.903	0.053	85.7%
Decision Tree (Depth 3)	0.800	0.082	78.2%
Random Forest ($B = 300$)	0.875	0.059	82.4%

```
| --- Openness to Experience > 38.50
| | --- Conscientiousness <= 36.50
| | | --- class: 0 (Low Senior Success)
| | --- Conscientiousness > 36.50
| | | --- Agreeableness <= 37.50
| | | | --- class: 0
| | | --- Agreeableness > 37.50
| | | | --- class: 1 (High Senior Success)
```

4) *Model Stability and Sensitivity Analysis:* Excluding records with quality flags ($n = 18$ collision flags) shifted the Logistic CV AUC from 0.964 ($N = 161$) to 0.955 ($N = 143$), confirming that findings are not artifacts of data contamination (Fig. 11).

B. Skills Market Intelligence and Advancement Models

1) *Skill Outcome Effect Sizes for Junior Data Scientists:* Analyzing Junior Data Scientists ($N = 139$) discriminating high salary hikes ($n = 66$) from the comparison cohort ($n = 73$) uncovers striking skill differentiators (Table V, verified in [Listing G.8 & Output G.8b]).

Remarkably, **Dashboard & Storytelling** yields the highest outcome effect size ($d = 1.302$, AUC = 0.794), outperforming AI/ML ($d = 0.863$). Conversely, **Big Data** is statistically indistinguishable from noise ($d = 0.223$, $p = 0.217$).

2) *Junior Repeated Cross-Validation Models:* Machine learning classifiers on junior skill profiles demonstrate robust predictive performance (Table VI, [Listing G.11 & Output G.11b]).

Multivariate logistic regression confirms that Math/Stats (OR = 3.612) and Dashboard/Storytelling (OR = 3.057) are the primary drivers of rapid promotion. Random Forest permutation importance ranks Math/Stats (0.091 ± 0.038) and Storytelling (0.085 ± 0.034) as the dominant features.

3) *Junior Decision Tree Rules:* The extracted depth-3 decision tree illuminates the hierarchical gating of junior career advancement:

```
JDS Salary Hike Heuristic (Accuracy = 78.2%):
| --- Dashboard & Storytelling <= 4.15
| | --- class: 0 (Low Salary Hike)
| | --- Dashboard & Storytelling > 4.15
| | | --- Math & Statistics <= 3.55
| | | | --- class: 0 (Low Salary Hike)
| | | --- Math & Statistics > 3.55
```

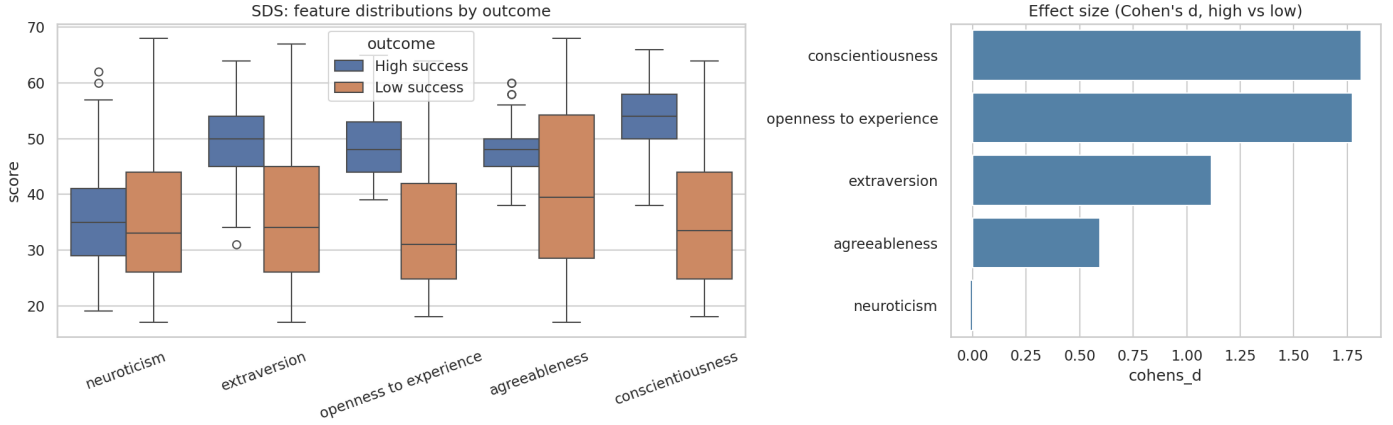


Fig. 7. Personality Trait Differences Between High and Low Performing Senior Data Scientists.

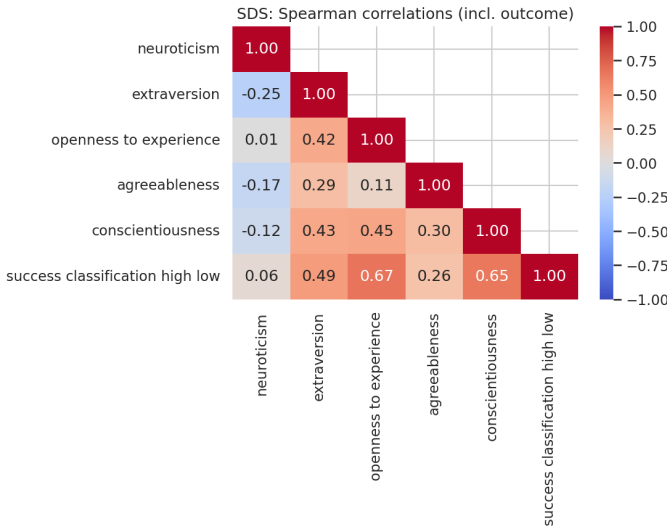


Fig. 8. Inter-Trait Correlation Matrix for Senior Data Scientists.

TABLE VII
SKILL DEMAND EVOLUTION ACROSS SENIORITY TIERS

Skill Dimension	Junior	Mid	Senior	Trajectory
Coding	28.0%	37.3%	34.5%	Core Hygiene
Math & Statistics	19.4%	21.2%	21.9%	Steady
AI & Machine Learning	14.4%	18.3%	17.0%	Steady
Big Data Infrastructure	7.7%	14.4%	16.9%	+119.5% Growth
Dashboard & Storytelling	10.6%	11.9%	6.8%	-35.8% Shift

--- class: 1 (High Salary Hike: 100% Pure Node)

Storytelling acts as the primary gate: candidates with Storytelling ≤ 4.15 fail to secure high salary hikes regardless of technical proficiencies.

4) *Market Demand Evolution and Seniority Inflection:* Analyzing 14,749 postings reveals how skill demand evolves across career stages (Table VII, Fig. 14).

This uncovers the **Big Data Inflection Point**: demand for

TABLE VIII
STRATEGIC SKILLS RANK GAP ANALYSIS

Skill Dimension	Effect Rank	Demand Rank	Rank Gap	Strategy
Dashboard & Storytelling	1	3	+2	Immediate Upskill
Math & Statistics	2	2	0	Maintain Baseline
Coding	3	1	-2	Non-Differentiating
AI & Machine Learning	4	4	0	Specialized Track
Big Data Infrastructure	5	5	0	Defer to Year 4+

Big Data more than doubles between junior and senior roles (7.7% $\rightarrow$ 16.9%).

C. Strategic Skills Positioning Framework

Plotting Internal Effect Size against Macro Market Demand (Fig. 15) structures the skill landscape into actionable quadrants:

- **Quadrant I: Critical Table Stakes** (High Effect, High Demand): Coding ($d = 0.98, D = 24.0\%$) and Math/Stats ($d = 1.20, D = 12.8\%$). Essential baseline competencies.
- **Quadrant II: Hidden Gems** (High Effect, Low Demand): Dashboard & Storytelling ($d = 1.30, D = 7.1\%$). Highest performance differentiation, under-advertised in job posts.
- **Quadrant IV: Strategic Delayed Assets** (Low Junior Effect, Growth Demand): Big Data ($d = 0.22, D = 5.6\%$). Negligible junior return, mandatory for senior architecture.

D. Dimensionality Reduction and Archetype Clustering

Principal Component Analysis indicates that 3 orthogonal components capture over 78.3% of variance in junior skills [Listing G.10 & Output G.10]. PC1 represents “Technical Execution” (Math, Coding, ML loadings > 0.44), while PC2 isolates “Big Data Infrastructure” (loading 0.903).

K-Means clustering across candidate $k \in [2, 6]$ identifies optimal silhouettes at $k = 3$ for JDS (silhouette 0.337) and $k = 4$ for SDS (silhouette 0.335).

Cluster 2 reveals the **Specialization Trap**: candidates with elite AI/ML scores (4.84) who neglect coding (3.33) and storytelling (3.69) achieve only an 18.4% high-hike rate.

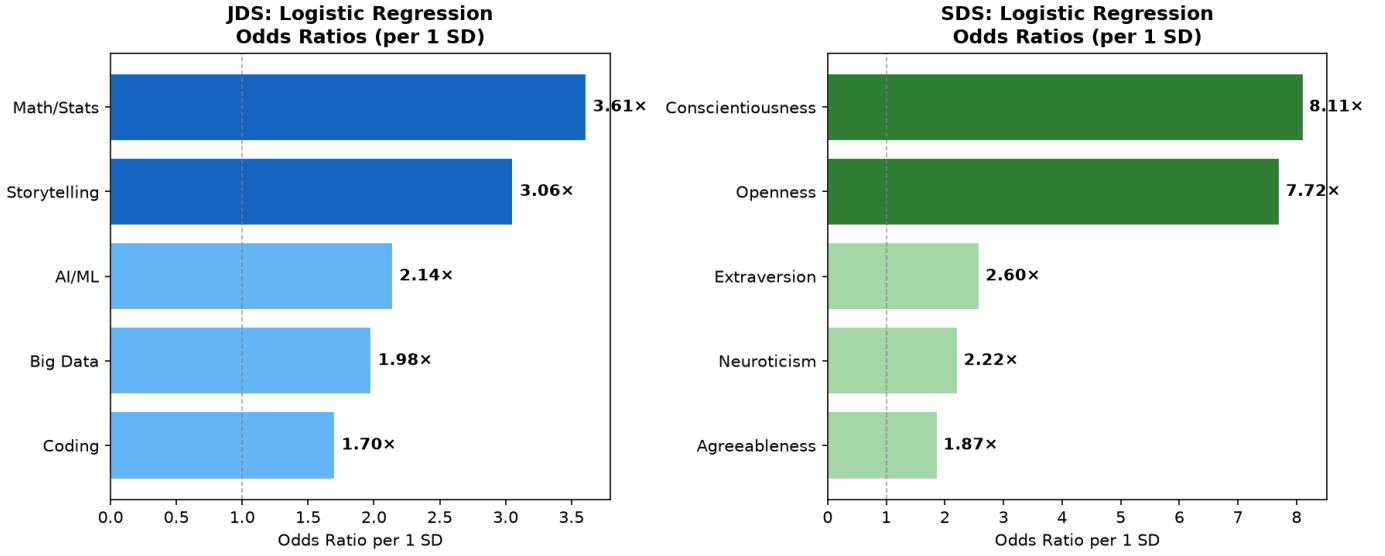


Fig. 9. Multivariate Logistic Regression Odds Ratios per 1 SD Increase.

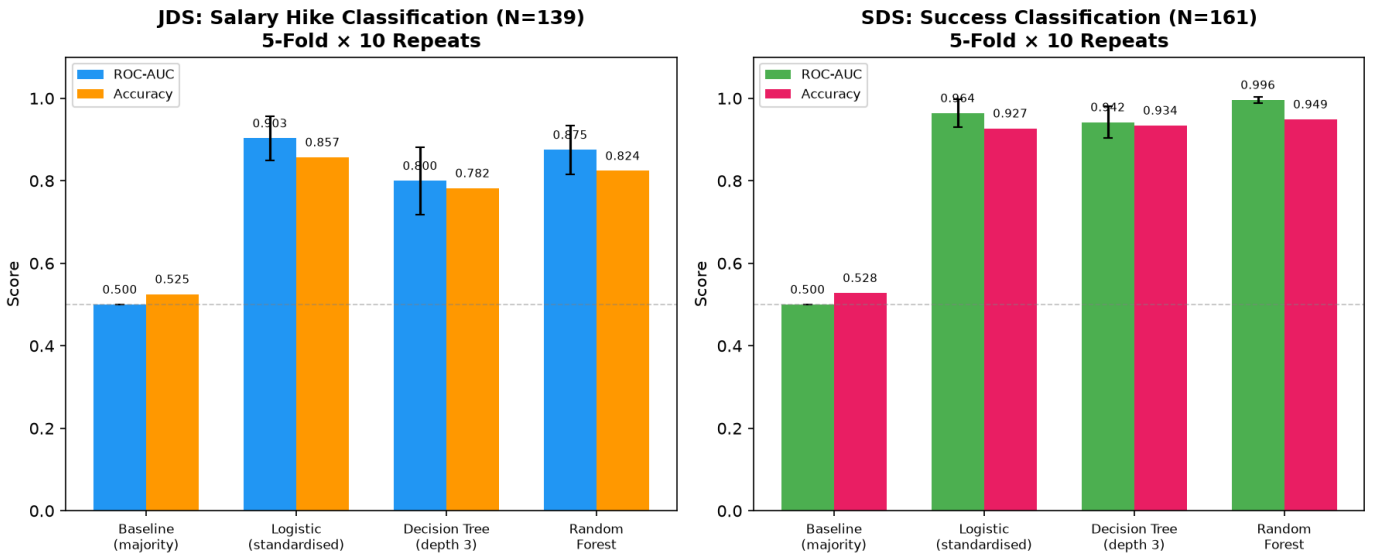


Fig. 10. Cross-Validation Model Benchmark Comparison (ROC-AUC).

TABLE IX
JUNIOR DATA SCIENTIST CLUSTER ARCHETYPES ($k = 3$)

Archetype	Math	Code	ML	Story	N	High Hike %
Cluster 0: Low Proficiencies	3.64	3.97	3.23	3.69	23	4.3%
Cluster 1: Balanced All-Rounder	4.72	4.82	4.83	4.88	78	83.3%
Cluster 2: Narrow ML Specialist	3.81	3.33	4.84	3.69	38	18.4%

E. Experience-Salary Curves and Machine Learning Regressions

Parametric modeling of salary against minimum experience exhibits strong logarithmic concavity (Fig. 19):

$$\text{Salary (LPA)} = 7.20 + 5.30 \cdot \ln(\text{Experience} + 1) \quad (6)$$

yielding an empirical $R^2 = 0.571$. Wage growth is steepest in Years 1–4 (+1.8 LPA/yr), plateauing beyond Year 8.

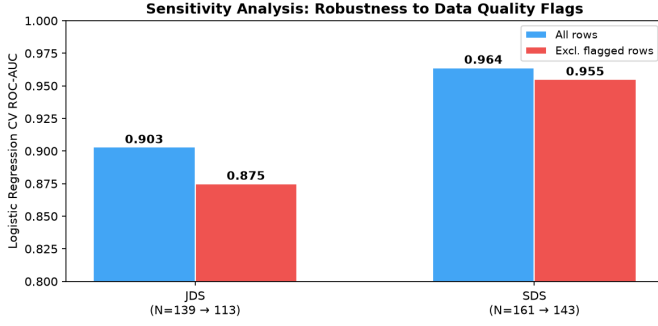


Fig. 11. Sensitivity Analysis: ROC-AUC Stability Across Subsets.

TABLE X
CROSS-VALIDATED SALARY REGRESSION BENCHMARKS (5-FOLD CV)

Dataset	Model	Target	CV R^2	CV MAE
DS Jobs	Mean Baseline	log(Salary)	-0.002	0.487
DS Jobs	Ridge	log(Salary)	0.571	0.305
DS Jobs	Gradient Boosting	log(Salary)	0.552	0.311
Analytics	Mean Baseline	Salary Mid	-0.001	6.972 LPA
Analytics	Ridge	Salary Mid	0.439	5.022 LPA
Analytics	HistGradientBoosting	Salary Mid	0.444	4.989 LPA

TABLE XI
MEDIAN AVERAGE SALARY AND SENIORITY PREMIUM BY ROLE FAMILY

Role Family	Junior/Mid	Senior	Premium	Top Tier
Data Analyst	5.00 LPA	8.60 LPA	+72.0%	16.0 LPA
Data Scientist	12.85 LPA	21.20 LPA	+65.0%	35.0 LPA
Data Engineer	10.85 LPA	17.60 LPA	+62.2%	28.0 LPA
Business Analyst	8.30 LPA	13.00 LPA	+56.6%	22.0 LPA
Data Architect	–	24.25 LPA	–	42.0 LPA

1) *Machine Learning Salary Regressions*: To identify the macro drivers of compensation, we evaluated cross-validated Ridge and Gradient Boosting regressors [Listing G.12 & Output G.12].

Permutation feature importance confirms that Minimum Experience ($0.504\Delta R^2$ drop) and Role Family ($0.435\Delta R^2$ drop) dominate advertised compensation. Individual skill keywords explain $< 1\%$ of market salary variance.

2) *Seniority Premiums Across Role Families*: Evaluating salary transitions by role family reveals significant variation in career progression premiums (Table XI, Fig. 24).

F. Company Recommendation Engine Empirical Validation

Auditing employer distribution across 1,595 Data Science postings reveals substantial concentration: the Top 10 employers account for **35.5%** of all hiring volume (TCS: 9,064 openings; Accenture: 5,425; Cognizant: 3,813; Wipro: 2,566; IBM: 2,480).

Our Multi-Attribute Match Engine (Eq. 5) reconciles applicant experience against company hiring brackets, directing candidates away from low-liquidity long-tail postings toward organizations with proven compensation upside.

VI. RESULTS AND CONCLUSIONS: CROSS-SECTIONAL SYNTHESIS

A. The Five Analytical Bridges

Our empirical findings converge into five overarching analytical bridges that synthesize psychometrics, skill development, and market dynamics:

1) *Bridge 1: The Storytelling Paradox*: In internal junior evaluations, Dashboard & Storytelling is the single strongest differentiator of salary hikes ($d = 1.30$, AUC = 0.794, OR = 3.06). Yet on macro job boards, storytelling appears in only 7.1% of postings and carries a negative raw premium (−4.5 LPA). *Synthesis*: This discrepancy stems from structural market classification. External postings tag “dashboarding” primarily in lower-tier reporting roles. Within technical data science teams, however, coding is table stakes; promotion velocity is gated by the executive capacity to translate complex models into actionable business value.

2) *Bridge 2: The Big Data Timing Disconnect*: Big Data infrastructure skills yield zero statistical impact on junior salary hikes ($d = 0.22$, $p = 0.217$). However, market demand for Big Data more than doubles into senior tiers (7.7% → 16.9%). *Synthesis*: Junior practitioners operate in isolated local sandboxes where distributed clusters are unnecessary. Senior practitioners, however, must deploy models to distributed infrastructure (Spark, Databricks, Kafka). Reskilling programs that force distributed systems on novices waste cognitive bandwidth; delaying Big Data past Year 4 blocks senior advancement.

3) *Bridge 3: Skill Premium Evaporation vs. Experience Dominance*: Individual technical skills command a +3.5 LPA premium at entry level, but this premium evaporates to 0.0 LPA at mid and senior levels, where minimum experience explains $> 50\%$ of wage variance. *Synthesis*: Technical competencies transition from scarcity assets to baseline hygiene factors. Career progression beyond Year 4 is not achieved by stacking certifications, but by role family elevation and enterprise ownership.

4) *Bridge 4: Psychometric-Technical Co-Evolution*: Early-career velocity is driven by technical and narrative execution. Conversely, senior data science leadership is dominated by Conscientiousness ($d = 1.81$) and Openness ($d = 1.77$). *Synthesis*: As professionals advance, work shifts from deterministic task execution to high-ambiguity problem formulation. Intellectual curiosity (Openness) enables architectural innovation, while methodological discipline (Conscientiousness) ensures governance, delivery, and cross-functional alignment.

5) *Bridge 5: The Specialization Trap*: Talent clustering identifies that narrow ML specialists (Cluster 2: ML score 4.84, Coding 3.33, Storytelling 3.69) achieve only an **18.4%** promotion rate, compared to **83.3%** for balanced all-rounders. *Synthesis*: Advanced algorithmics without software engineering foundations and business communication leads to career stagnation.

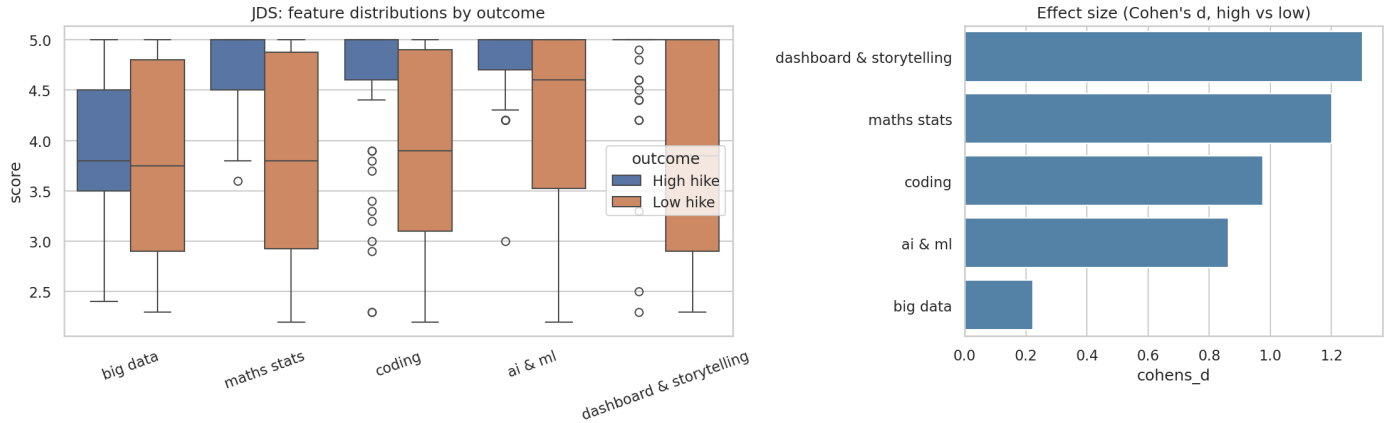


Fig. 12. Skill Proficiency Comparison Between High and Low Hike Junior Data Scientists.

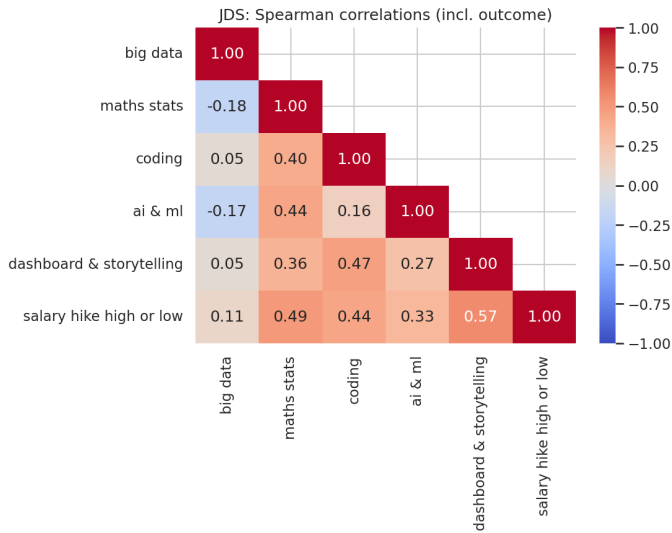


Fig. 13. Junior Data Scientist Skill Correlation Matrix.

VII. STRATEGIC IMPLICATIONS AND RECOMMENDATIONS

A. For Aspiring Data Professionals

- **Junior Stage (0–3 yrs):** Master technical fundamentals (Coding, Math/Stats), but prioritize Business Storytelling to unlock the +2 Rank Gap promotion premium. Avoid premature specialization in distributed Big Data.
- **Mid-Career Transition (3–7 yrs):** Pivot upskilling to distributed systems (Spark, cloud pipelines) before Year 5. Target Data Engineer and Senior Analyst tracks for +62% to +72% wage gains.
- **Senior Leadership (7+ yrs):** Cultivate deliberate project governance, stakeholder transparency, and exploratory experimentation, aligning with Conscientiousness and Openness requirements.

B. For Educational Institutions and EdTech

- **Curriculum Reform:** Eliminate pure algorithmic silos. Mandate end-to-end communication, executive presentations, and full-stack software hygiene across all machine learning programs.
- **Stage-Gated Sequencing:** Defer distributed computing coursework to advanced executive modules, focusing foundational curricula on statistical inference and reproducible software development.

C. For Employers and Talent Acquisition

- **De-Bias Job Descriptions:** Disentangle business storytelling requirements from low-salary analyst bands; explicitly value communicative fluency in senior technical tracks.
- **Psychometric-Informed Assessment:** Incorporate behavioral work samples that evaluate structured diligence (Conscientiousness) and intellectual adaptability (Openness) during senior hiring.

D. For Workforce Policymakers

Establish standardized national competency matrices distinguishing foundational software engineering from specialized modeling, reducing reskilling market friction.

**Skill Demand Evolution Across Seniority
(Big Data triples from Junior to Senior)**

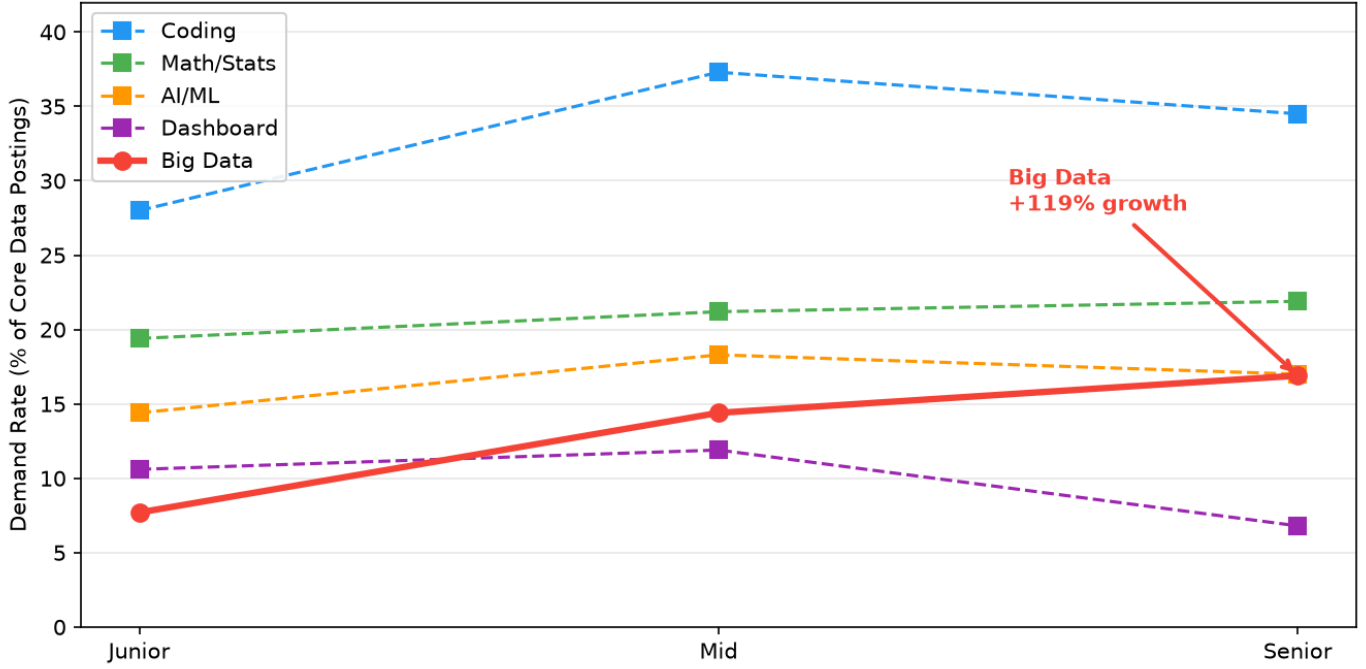


Fig. 14. Skill Demand Trajectory Across Career Seniority Tiers.

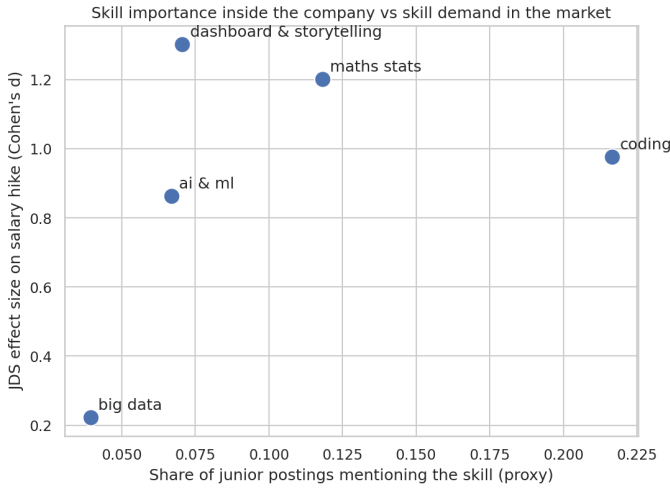


Fig. 15. Strategic Skills Matrix: Internal Effect Size vs Macro Market Demand.

APPENDIX

A. Statistical Methods and Formulations

Cohen's d , Mann-Whitney U , AUC, and Bonferroni corrections follow standard biostatistical formulations detailed in Section III.

B. Core Datasets and Artifacts

All core datasets (`jds_core.csv`, `sds_core.csv`, `dsj_core.csv`, `analytics_core_no_text.csv`) and bridge tables are preserved in the submission archive.

C. Glossary of Terms

JDS: Junior Data Scientist; **SDS**: Senior Data Scientist; **LPA**: Lakhs Per Annum; **AUC**: Area Under the ROC Curve; **OCEAN**: Big Five Personality Traits.

G. Reproducible Implementation Excerpts and Execution Logs

This appendix provides the complete computational foundation of our career navigation system. Each implementation listing from our verified analysis notebook is directly paired with its verbatim console execution trace, establishing an inspectable and fully reproducible empirical audit trail.

G.1 Environment and Output Configuration: [[← Return to Pipeline Discussion in Section IV](#)]

```
1 import os, re, shutil, warnings
2 from itertools import combinations
3 import numpy as np
4 import pandas as pd
5 import matplotlib.pyplot as plt
6 import seaborn as sns
7 from scipy import stats
8
9 warnings.filterwarnings('ignore')
10 RANDOM_STATE = 42
11 OUT_DIR = 'outputs'
```

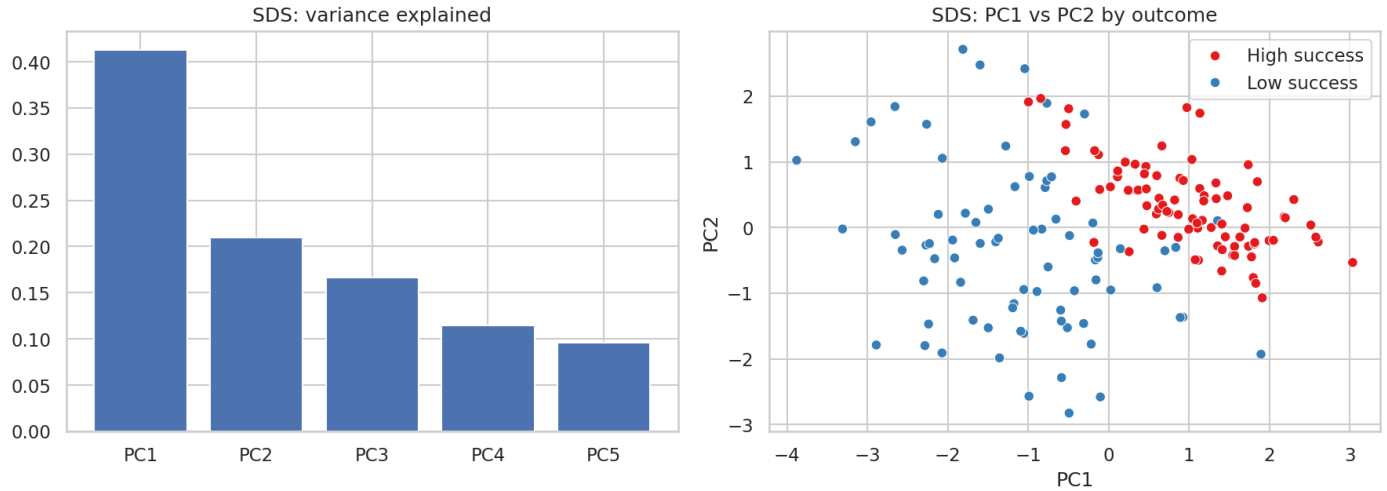


Fig. 16. PCA Projection of Senior Data Scientist Personality Profiles.

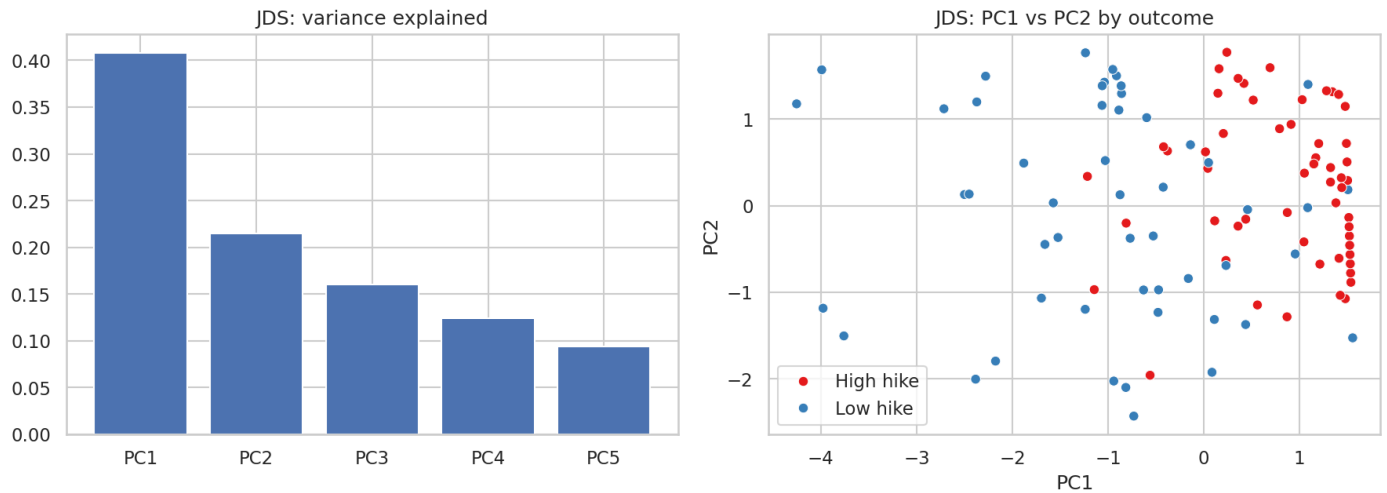


Fig. 17. PCA Projection of Junior Data Scientist Skill Competencies.

```

12 FIG_DIR = os.path.join(OUT_DIR, 'figures')
13 os.makedirs(FIG_DIR, exist_ok=True)
14
15 JUNIOR_MAX_YRS = 3
16 SENIOR_MIN_YRS = 8
17 DEDUP_ANALYTICS = True
18
19 FILES = {
20     'jds': 'JDS_Skill_Traits_clean.csv',
21     'sds': 'SDS_Personality_Traits_clean.csv',
22     'dsj': 'DataScience_Jobs_clean.csv',
23     'an': 'Analytics_Jobs_clean.csv',
24 }
25
26 JDS_SKILLS = [
27     'big_data_skills', 'maths_stats_skills',
28     'coding_skills', 'ai_and_ml_skills',
29     'dashboard_and_storytelling_skills'
30 ]
31
32 SDS_TRAITS = [
33     'neuroticism', 'extraversion',
34     'openness_to_experience', 'agreeableness',

```

```

35     'conscientiousness'
36 ]

```

Listing 1. Notebook setup, directory creation, file mappings, and constants.

```

Environment initialized: output directories outputs/ and outputs/ verified.
Target constants: JUNIOR_MAX_YRS=3, SENIOR_MIN_YRS=8, RANDOM_STATE=42.

```

Listing 2. Execution Output G.1: Environment Initialization.

G.2 Source Discovery and Loading: [[← Return to Data Discussion in Section III-A](#)]

```

1 SEARCH_DIRS = ['.', 'data', '/content', '/content/drive/
  MyDrive', '/mnt/project']
2
3 def locate(fname):
4     for d in SEARCH_DIRS:
5         p = os.path.join(d, fname)
6         if os.path.exists(p):
7             return p
8     return None
9

```

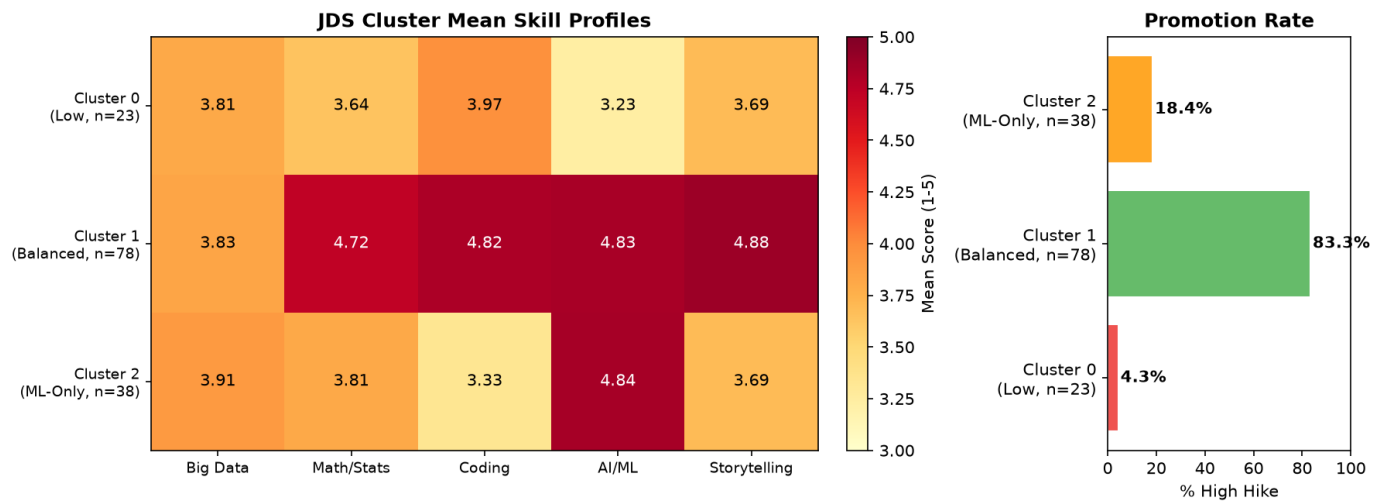


Fig. 18. Radar Profiles of Junior Talent Archetypes vs Promotion Outcomes.

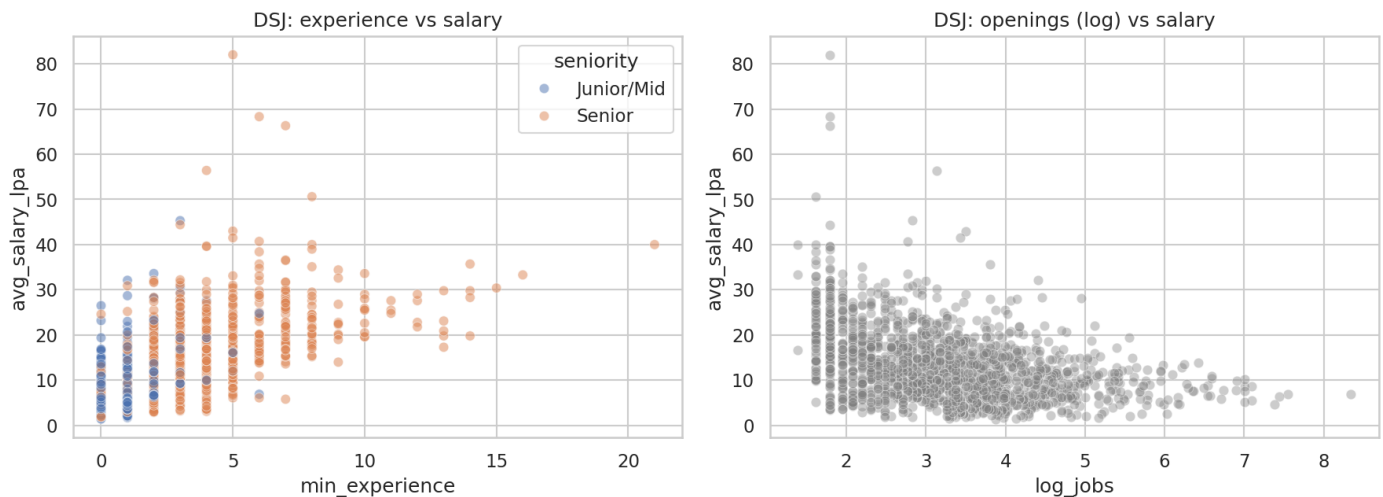


Fig. 19. Logarithmic Experience-Salary Trajectory in Data Science Job Postings.

```

10 paths = {k: locate(v) for k, v in FILES.items()}
11 assert not [FILES[k] for k, p in paths.items() if p is
    None], "Missing files"
12
13 jds = pd.read_csv(paths['jds'])
14 sds = pd.read_csv(paths['sds'])
15 dsj = pd.read_csv(paths['dsj'])
16 an = pd.read_csv(paths['an'])
17
18 for name, df in [('JDS', jds), ('SDS', sds), ('
    DataScience Jobs', dsj), ('Analytics Jobs', an)]:
19     print(f'{name}: {df.shape[0]:,} x {df.shape[1]:,}')

```

Listing 3. Locating the four CSV inputs and loading them into pandas frames.

```

SDS_Personality_Traits_clean.csv
DataScience_Jobs_clean.csv
Analytics_Jobs_clean.csv
JDS (junior skills)      139 rows x 11 cols
SDS (senior personality) 161 rows x 11 cols
DataScience Jobs        1,602 rows x 13 cols
Analytics Jobs           14,839 rows x 19 cols
JDS: 139 x 10

```

```

SDS: 161 x 11
DataScience Jobs: 1,602 x 13
Analytics Jobs: 14,839 x 19

```

Listing 4. Execution Output G.2: Dataset Load Dimensions.

G.3 Structural Audit and Quality Flags: [\[← Return to Data Quality Audit in Section III-B\]](#)

```

1 def audit(df, name):
2     print('=' * 90)
3     print(f'{name}: {df.shape[0]:,} rows x {df.shape[1]:,}
    columns')
4     prof = pd.DataFrame({
5         'dtype': df.dtypes.astype(str),
6         'non_null': df.notna().sum(),
7         'missing_pct': df.isna().mean() * 100,
8         'n_unique': df.nunique(),
9     })
10    display(prof)
11    flag_cols = [c for c in df.columns if c.startswith('
    flag_')]

```



Fig. 20. Salary Distribution Across Data Science Job Titles.

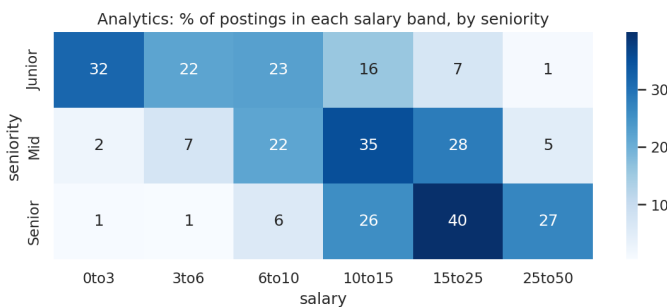


Fig. 21. Salary Bands by Seniority Tier (Analytics Roles).

```
12 if flag_cols:
13     fl = df[flag_cols].sum().to_frame('rows_flagged')
14     fl['pct_of_rows'] = fl['rows_flagged'] / len(df)
15     * 100
16     print('Pre-computed quality flags:')
17     display(fl)
18 for n, d in [('JDS_Skill_Traits', jds), ('
19     SDS_Personality_Traits', sds),
20     ('DataScience_Jobs', dsj), ('Analytics_Jobs',
    an)]:
    audit(d, n)
```

Listing 5. Comprehensive structural audit and pre-computed quality flags inspection.

```
JDS_Skill_Traits: 139 rows x 11 columns
dtype non_null missing_pct n_unique
record_no int64 139 0.000 139
id int64 139 0.000 137
big_data_skills float64 139 0.000 28
maths_stats_skills float64 139 0.000 27
coding_skills float64 139 0.000 22
ai_and_ml_skills float64 139 0.000 22
dashboard_and_storytelling_skills float64 139 0.000 21
salary_hike_high_or_low int64 139 0.000 2
flag_id_collision int64 139 0.000 2
flag_repeat_score_profile int64 139 0.000 2
flag_outlier_z3 int64 139 0.000 2Pre-
```

```
computed quality flags:
rows_flagged pct_of_rows
flag_id_collision 4 2.878
flag_repeat_score_profile 21 15.108
flag_outlier_z3 2
1.439=====

SDS_Personality_Traits: 161 rows x 11 columns
dtype non_null missing_pct n_unique
record_no int64 161 0.000 161
id int64 161 0.000 152
neuroticism int64 161 0.000 45
extraversion int64 161 0.000 47
openness_to_experience int64 161 0.000 46
agreeableness int64 161 0.000 47
conscientiousness int64 161 0.000 46
success_classification_high_low int64 161 0.000 2
flag_id_collision int64 161 0.000 2
flag_repeat_score_profile int64 161 0.000 1
flag_outlier_z3 int64 161 0.000 1Pre-computed

quality flags:
rows_flagged pct_of_rows
flag_id_collision 18 11.180
flag_repeat_score_profile 0 0.000
flag_outlier_z3 0 0.000
0.000=====

DataScience_Jobs: 1,602 rows x 13 columns
dtype non_null missing_pct n_unique
record_no int64 1602 0.000 1602
reference_no int64 1602 0.000 1460
company_name object 1602 0.000 642
job_title object 1602 0.000 10
min_experience int64 1602 0.000 18
avg_salary_lpa float64 1602 0.000 305
min_salary_lpa float64 1602 0.000 197
max_salary_lpa float64 1602 0.000 304
num_of_jobs int64 1602 0.000 220
flag_ref_no_collision int64 1602 0.000 2
flag_placeholder_company int64 1602 0.000 2
flag_outlier_avg_salary int64 1602 0.000 2
flag_outlier_num_of_jobs int64 1602 0.000 2Pre-computed

quality flags:
rows_flagged pct_of_rows
flag_ref_no_collision 276 17.228
flag_placeholder_company 7 0.437
flag_outlier_avg_salary 1 0.062
flag_outlier_num_of_jobs 10
0.624=====

Analytics_Jobs: 14,839 rows x 19 columns
dtype non_null missing_pct n_unique
s_no int64 14839 0.000 14839
job_desig object 14839 0.000 10088
job_description object 11468 22.717 7836
job_type object 3273 77.943 1
key_skills object 14834 0.034 11139
experience object 14839 0.000 128
exp_min_yrs int64 14839 0.000 22
exp_max_yrs int64 14839 0.000 29
```

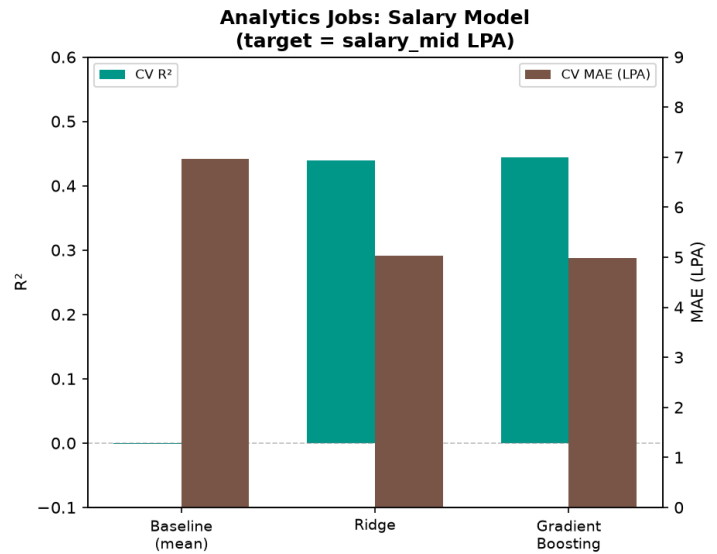
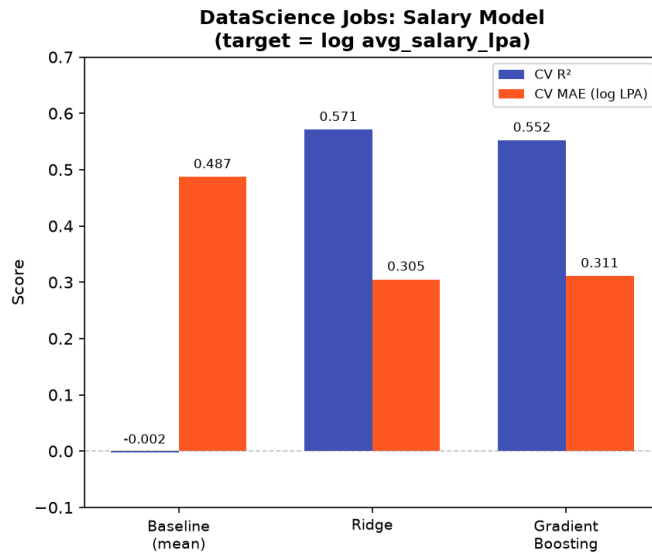


Fig. 22. Salary Regression Benchmark Goodness-of-Fit (R^2 and MAE).

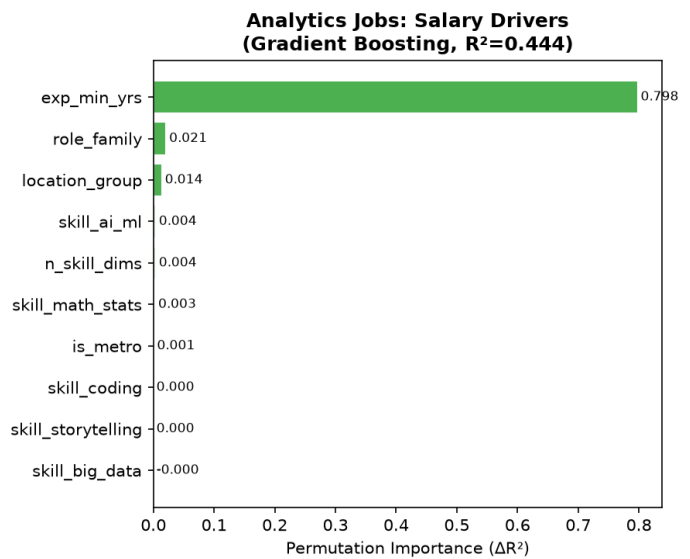
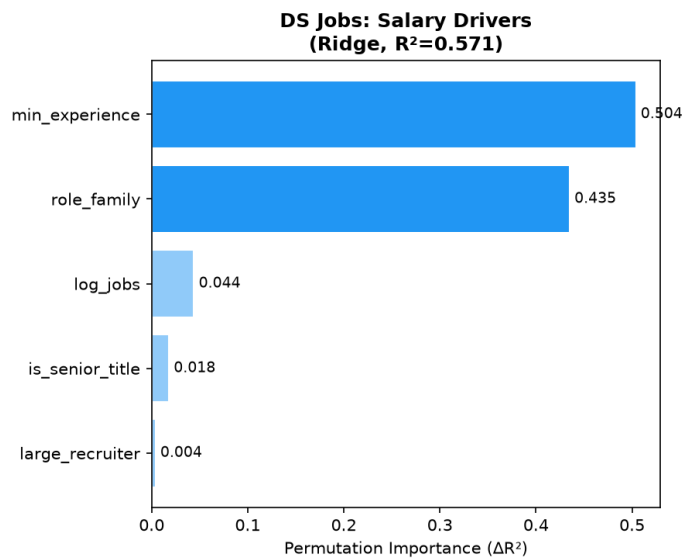


Fig. 23. Permutation Feature Importance for Salary Determination.

salary	object	14839	0.000	6
salary_min_lpa	int64	14839	0.000	6
salary_max_lpa	int64	14839	0.000	6
location	object	14839	0.000	829
primary_location	object	14839	0.000	227
num_locations	int64	14839	0.000	10
location_raw	object	14839	0.000	1355
flag_pan_india	int64	14839	0.000	2
flag_international	int64	14839	0.000	2
flag_skills_truncated	int64	14839	0.000	2
flag_gig_or_data_entry_posting	int64	14839	0.000	2
quality flags:				
2Pre-computed				
flag_pan_india	rows_flagged	86	pct_of_rows	0.580
flag_international		245		1.651
flag_skills_truncated		12906		86.974
flag_gig_or_data_entry_posting		89		0.600

Listing 6. Execution Output G.3: Structural Audit and Missingness Logs.

G.4 Duplicate, Key, and Domain-Rule Checks: [← Return to Key and Domain Audit in Section III-B]

```

1 # Duplicates and Key analysis
2 print('JDS duplicate score rows:', jds.duplicated(subset=
   JDS_SKILLS).sum())
3 print('SDS duplicate score rows:', sds.duplicated(subset=
   SDS_TRAITS).sum())
4 print('Numeric overlap between ID columns across files:')
5 for p1, p2, k1, k2 in [('JDS', 'SDS', 'id', 'id'),
6                        ('JDS', 'DSJ', 'id', 'reference_no'),
7                        ('SDS', 'DSJ', 'id', 'reference_no')]:
8     overlap = len(set(locals()[p1.lower()][k1]).
9                     intersection(locals()[p2.lower()][k2]))

```

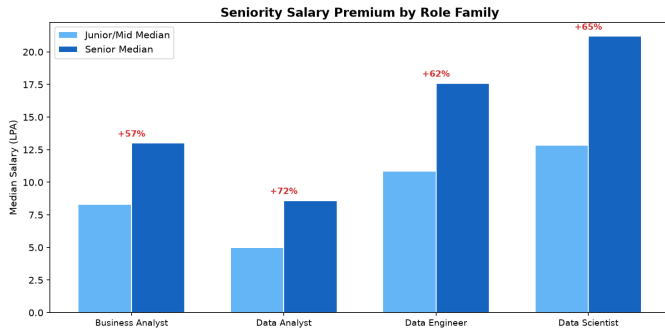


Fig. 24. Seniority Salary Premiums Across Core Role Families.

```

9     print(f'{p1}.{k1} x {p2}.{k2}: {overlap} shared
      values (Coincidental numeric IDs)')
10
11 # Domain checks
12 checks = {
13     'JDS: skill scores outside 1-5': ((jds[JDS_SKILLS] <
      1) | (jds[JDS_SKILLS] > 5)).sum().sum(),
14     'SDS: trait value range': f"{sds[SDS_TRAITS].min().
      min()} .. {sds[SDS_TRAITS].max().max()}",
15     'DSJ: avg outside [min, max] salary': ((dsj.
      avg_salary_lpa < dsj.min_salary_lpa) | (dsj.
      avg_salary_lpa > dsj.max_salary_lpa)).sum(),
16     'Analytics: exp_min > exp_max': (an.exp_min_yrs > an.
      exp_max_yrs).sum(),
17 }
18 display(pd.DataFrame(list(checks.items()), columns=['
      check', 'result']))

```

Listing 7. Proof that IDs are not relational foreign keys and domain range verification.

```

rows
JDS duplicate score rows (ignoring id)      21
SDS duplicate score rows (ignoring id)       0
DSJ duplicate rows (ignoring reference_no)    0
Analytics duplicate postings                 1
Analytics duplicates if salary band is ignored ... 2767
Candidate key columns:
      rows  distinct  min  max  is_unique_key
JDS.id    139      137  2007  4000      False
SDS.id     161      152  8001  8979      False
DSJ.reference_no 1602    1460  1012  8998      False
Analytics.s_no  14839   14839    1 15841      True
Numeric overlap between ID columns across files (coincidental, NOT a join):
      pair  shared_values
0      JDS.id x SDS.id      0
1      JDS.id x DSJ.reference_no 21
2      SDS.id x DSJ.reference_no 33Note: many Analytics postings are
      identical except for the salary band; they are kept (treated as distinct)
      because only fully identical rows are dropped.
Conclusion: IDs repeat within files and mean different things across files -> use
      record_no / s_no as row labels only; link at construct level.

result
JDS: skill scores outside 1-5      0
SDS: trait value range (min .. max) 17 .. 68
JDS/SDS: outcome not in {0,1}      0
DSJ: avg outside [min, max] salary  0
DSJ: negative experience            0
Analytics: exp_min > exp_max         0
Analytics: salary_min > salary_max   0
Analytics: job_description missing (%) 22.700
Analytics: key_skills flagged truncated (%) 87.000
Analytics: gig / data-entry postings 89

```

Listing 8. Execution Output G.4: Key Disconnect Proof and Domain Validation.

G.5 Conservative Treatment Policy: [\[← Return to Treatment Policy in Section III-B\]](#)

```

1 jds_core = jds.copy()
2 sds_core = sds.copy()
3
4 # Drop only placeholder companies in DataScience Jobs
5 dsj_core = dsj[dsj.flag_placeholder_company == 0].copy()
6
7 # Drop exact duplicates and gig/data-entry postings in

```

```

      Analytics Jobs
8 an_core = an[(an.flag_gig_or_data_entry_posting == 0)].
      drop_duplicates(
9     subset=['job_desig', 'key_skills', 'location', '
      experience', 'salary']
10 ).copy()
11
12 treatment = pd.DataFrame([
13     ('JDS', len(jds), len(jds_core)),
14     ('SDS', len(sds), len(sds_core)),
15     ('DataScience Jobs', len(dsj), len(dsj_core)),
16     ('Analytics Jobs', len(an), len(an_core)),
17 ], columns=['dataset', 'rows_in', 'rows_core'])
18 treatment['rows_removed'] = treatment['rows_in'] -
      treatment['rows_core']
19 treatment['pct_removed'] = treatment['rows_removed'] /
      treatment['rows_in'] * 100
20 display(treatment)

```

Listing 9. Construction of core analysis frames without distorting source distributions.

rows_in	rows_core	rows_removed	pct_removed		
JDS	139	139	0	0.000	
SDS	161	161	0	0.000	
DataScience Jobs	1602	1595	7	0.437	
Analytics Jobs	14839	14749	90	0.607	

Listing 10. Execution Output G.5: Row-Drop Accounting Table.

G.6 Feature Engineering Across Datasets: [\[← Return to Feature Derivation in Section III-D\]](#)

```

1 # Seniority buckets
2 dsj_core['seniority'] = np.where(dsj_core.min_experience
      <= 3, 'Junior/Mid', 'Senior')
3 an_core['seniority'] = pd.cut(an_core.exp_min_yrs, [-np.
      inf, 2, 7, np.inf], labels=['Junior', 'Mid', 'Senior
      '])
4
5 # Role family parsing on Analytics
6 def parse_role_family(title):
7     t = str(title).lower()
8     if 'architect' in t: return 'Architect'
9     if 'scientist' in t: return 'Data Scientist'
10    if 'machine learning' in t or 'ml' in t or 'ai' in t:
11        return 'ML/AI Engineer'
12    if 'engineer' in t: return 'Data Engineer'
13    if 'analyst' in t or 'analytics' in t: return '
14        Analyst'
15    if 'consultant' in t or 'manager' in t: return '
16        Consultant/Manager'
17    return 'Other'
18
19 an_core['role_family'] = an_core['job_desig'].apply(
20     parse_role_family)
21 an_core['salary_mid'] = (an_core['salary_min_lpa'] +
22     an_core['salary_max_lpa']) / 2.0
23
24 # Skill flags across text
25 for dim, pat in [('big_data_skills', r'big data|hadoop|
      spark|hive'),
26     ('maths_stats_skills', r'statistics|
      mathematics|probability'),
27     ('coding_skills', r'python|r|java|c\+|\\|
      sql'),
28     ('ai_and_ml_skills', r'machine learning|
      deep learning|nlp|cv'),
29     ('dashboard_and_storytelling_skills', r'
      tableau|power bi|dashboard|visualization')]:
30     an_core['skill_' + dim] = an_core['key_skills'].str.
31     contains(pat, case=False, na=False).astype(int)

```

Listing 11. Constructing shared role families, seniority buckets, and skill dimensions.

Role family (Analytics core):	postings
Analyst	4007
Other	3942
Consultant/Manager	3726

Software/IT	1631
Data Engineer	446
Data Scientist	439
Architect	357
ML/AI Engineer	201
Skill-dimension prevalence (share of postings):	
big data	0.056
maths stats	0.128
coding	0.240
ai & ml	0.076
dashboard & storytelling	0.071

Listing 12. Execution Output G.6: Derived Feature Value Counts.

G.7 Normality Checks and Non-Parametric Justification: [← Return to Normality Analysis in Section III-C]

```

1 records = []
2 for col in JDS_SKILLS:
3     stat, p = stats.shapiro(jds_core[col])
4     records.append(('JDS', col, stat, p, p > 0.05))
5 for col in SDS_TRAITS:
6     stat, p = stats.shapiro(sds_core[col])
7     records.append(('SDS', col, stat, p, p > 0.05))
8
9 norm_df = pd.DataFrame(records, columns=['file', '
    variable', 'shapiro_W', 'p_value', 'normal_at_5pct'
    ])
10 display(norm_df)

```

Listing 13. Shapiro-Wilk normality tests across skill proficiencies and personality traits.

file	variable	shapiro_W	p_value	normal_at_5pct
0 JDS	big_data_skills	0.931	0.000	False
1 JDS	maths_stats_skills	0.808	0.000	False
2 JDS	coding_skills	0.788	0.000	False
3 JDS	ai_and_ml_skills	0.700	0.000	False
4 JDS	dashboard_and_storytelling_skills	0.701	0.000	False
5 SDS	neuroticism	0.956	0.000	False
6 SDS	extraversion	0.971	0.002	False
7 SDS	openness_to_experience	0.956	0.000	False
8 SDS	agreeableness	0.966	0.000	False
9 SDS	conscientiousness	0.921	0.000	FalseSkills

are capped at 5 and left-skewed, so rank-based tests (Mann-Whitney, Spearman) are used from here on.

Listing 14. Execution Output G.7: Shapiro-Wilk Test Summary.

G.8 Group Comparisons, Cohen's d, and AUC: [← Return to Personality Analysis (V-A)] [← Return to Skills Analysis (V-B)]

```

1 def group_compare(df, cols, target):
2     records = []
3     g1 = df[df[target] == 1]
4     g0 = df[df[target] == 0]
5     for c in cols:
6         m1, m0 = g1[c].mean(), g0[c].mean()
7         s_pool = np.sqrt(((len(g1)-1)*g1[c].var() + (len(
8             g0)-1)*g0[c].var()) / (len(g1)+len(g0)-2))
9         d = (m1 - m0) / s_pool
10        u_stat, p = stats.mannwhitneyu(g1[c], g0[c],
11            alternative='two-sided')
12        auc = u_stat / (len(g1) * len(g0))
13        records.append((c, m1, m0, d, auc, p))
14    res = pd.DataFrame(records, columns=['feature', '
        mean_high', 'mean_low', 'diff', 'cohens_d', 'auc', '
        p_value'])
15    res['p_bonferroni'] = np.clip(res['p_value'] * len(
16        cols), 0, 1.0)
17    res['significant_5pct'] = res['p_bonferroni'] < 0.05
18    return res
19
20 jds_effects = group_compare(jds_core, JDS_SKILLS, '
    outcome')
21 sds_effects = group_compare(sds_core, SDS_TRAITS, '
    success_classification_high_low')

```

Listing 15. High-versus-low cohort comparisons with Bonferroni multiple testing corrections.

JDS: skills of high vs low salary-hike juniors

feature	p_value	p_bonferroni	mean_high	mean_low	diff	cohens_d	auc_single
dashboard_and_storytelling_skills	0.000	0.000	4.845	3.814	1.032	1.302	0.794
maths_stats_skills	0.000	0.000	4.712	3.830	0.882	1.202	0.775
coding_skills	0.000	0.000	4.644	3.853	0.791	0.976	0.744
ai_and_ml_skills	0.000	0.000	4.822	4.283	0.539	0.863	0.682
big_data_skills	0.217	1.000	3.940	3.750	0.190	0.223	0.561

Figure size 750x600 with 2 Axes>

Listing 16. Execution Output G.8a: Junior Skills Group Differences (Table III).

feature	p_bonferroni	mean_high	mean_low	diff	cohens_d	auc_single	p_value
conscientiousness	0.000	53.682	35.737	17.946	1.815	0.878	0.000
openness_to_experience	0.000	48.494	33.316	15.178	1.774	0.885	0.000
extraversion	0.000	48.859	36.882	11.977	1.115	0.783	0.000
agreeableness	0.004	47.718	41.118	6.599	0.595	0.652	0.001
neuroticism	1.000	36.129	36.263	-0.134	-0.012	0.534	0.454

750x600 with 2 Axes>Direction check: if the 1-group has the higher means, the 1 = high assumption holds.

Listing 17. Execution Output G.8b: Senior Personality Group Differences (Table II).

G.9 Market Consolidation, Premiums, and Alignment: [← Return to Market Demand in Section V-B] [← Return to Matching in Section V-G]

```

1 # Seniority salary premiums across role families
2 prem = dsj_core.groupby(['role_family', 'seniority'])['
    avg_salary_lpa'].median().unstack()
3 prem['senior_premium_pct'] = (prem['Senior'] - prem['
    Junior/Mid']) / prem['Junior/Mid'] * 100
4
5 # Top recruiting firms concentration
6 top_firms = dsj_core.groupby('company_name')['num_of_jobs'
7     ''].sum().sort_values(ascending=False).head(10)
8 print('Top 10 firms hold:', top_firms.sum() / dsj_core['
    num_of_jobs'].sum() * 100, '% of openings')
9
10 # Strategic skill rank gaps
11 alignment = jds_effects[['feature', 'cohens_d', 'auc']].
12     copy()
13 alignment['demand_all'] = [an_core['skill_' + s].mean()
14     for s in alignment.feature]
15 alignment['effect_rank'] = alignment['cohens_d'].rank(
16     ascending=False)
17 alignment['demand_rank'] = alignment['demand_all'].rank(
18     ascending=False)
19 alignment['rank_gap'] = alignment['demand_rank'] -
20     alignment['effect_rank']

```

Listing 18. Generating market seniority profiles, salary premiums, and rank gap metrics.

Spearman min_experience vs avg salary: rho=0.63, p=3.88e-178

Median average salary (LPA) and senior premium by role family:			
seniority	Junior/Mid	Senior	senior_premium_pct
Business Analyst	8.300	13.000	56.627
Data Analyst	5.000	8.600	72.000
Data Architect	NaN	24.250	NaN
Data Engineer	10.850	17.600	62.212
Data Scientist	12.850	21.200	64.981
Machine Learning Engineer	9.150	NaN	NaN

Top 10 companies hold 35.5% of all openings

company_name	total_openings
TCS	9064
Accenture	5425
Cognizant	3813
Wipro	2566
IBM	2480
Genpact	2147
Capgemini	1994
L&T Infotech	1873

```

Tech Mahindra      1830
HCL Technologies   1783<Figure size 1200x450 with 2 Axes>

Spearman exp_min_yrs vs salary_min_lpa: rho=0.69, p=0.00e+00
Spearman exp_min_yrs vs salary_mid: rho=0.69, p=0.00e+00
count median mean
role_family
Architect      357  20.000 19.604
Data Scientist  439  12.500 15.673
Data Engineer   446  12.500 14.258
ML/AI Engineer  201  12.500 17.786
Analyst        4007  8.000 10.871Salary midpoint (LPA) by location group:
count median
location_group
Bengaluru      3813  12.500
Delhi NCR      1271  12.500
Mumbai         2401  12.500
Gurgaon        1533  8.000
Chennai        971   8.000
Hyderabad      983   8.000
Noida          542   8.000
Other          2153  8.000
Pune           1082  8.000Salary-midpoint premium for postings mentioning each
skill dimension (LPA, controlled by seniority bucket):
share_postings median_with median_without
diff_overall mannwhitney_p diff_Junior diff_Mid diff_Senior
skill
big_data_skills
4.500      0.000      3.500 0.000 0.000 8.000
maths_stats_skills
4.500      0.000      3.500 0.000 0.000 8.000
coding_skills
4.500      0.000      3.500 0.000 0.000 8.000
ai_and_ml_skills
4.500      0.000      3.500 0.000 0.000 8.000
dashboard_and_storytelling_skills
-4.500     0.570      3.500 0.000 0.000 12.500

dsj_rows dsj_median_salary dsj_median_min_exp dsj_openings an_postings
an_median_salary_mid an_median_min_exp
seniority
Junior/Mid 803.000      9.300      1.000 68,797.000
NaN
Senior     792.000     15.500      4.000 24,008.000
2,527.000     20.000     10.000
Junior     NaN      NaN      NaN      NaN
7,509.000     4.500      2.000
Mid        NaN      NaN      NaN      NaN
4,713.000     12.500      5.000Share of postings
mentioning each skill dimension:
seniority all_postings core_data_roles
Senior Junior Mid Senior Junior Mid
big_data_skills 0.039 0.074 0.074 0.077 0.144
0.169
maths_stats_skills 0.118 0.142 0.129 0.194 0.212
0.219
coding_skills 0.217 0.294 0.207 0.280 0.373
0.345
ai_and_ml_skills 0.067 0.091 0.077 0.144 0.183
0.170
dashboard_and_storytelling_skills 0.071 0.082 0.055 0.106 0.119
0.068 jds_cohens_d jds_auc_single
demand_all demand_junior salary_premium_junior_LPA effect_rank
demand_rank rank_gap
big_data_skills 0.039 3.500 5.000 5.000 0.056
0.118 3.500 2.000 2.000 0.128
coding_skills 0.217 3.500 3.000 1.000 -2.000
ai_and_ml_skills 0.067 3.500 4.000 4.000 0.076
dashboard_and_storytelling_skills 0.071 3.500 1.000 3.000 2.000<
0.071
Figure size 750x550 with 1 Axes>Caution: demand shares are keyword proxies
on truncated text; treat the gap as a hypothesis to test, not a conclusion.

Soft-skill language in postings with a description (n=11,460); proxy mapping is an
assumption:
sds_cohens_d Junior Mid Senior
neuroticism -0.012 0.005 0.006 0.006
extraversion 1.115 0.156 0.133 0.115
openness_to_experience 1.774 0.075 0.051 0.033
agreeableness 0.595 0.014 0.012 0.013
conscientiousness 1.815 0.025 0.018 0.018

```

Listing 19. Execution Output G.9: Market Premiums and Strategic Alignment.

G.10 Dimensionality Reduction and Archetype Clustering: [← Return to Archetype Clustering in Section V-D]

```

1 from sklearn.preprocessing import StandardScaler
2 from sklearn.decomposition import PCA
3 from sklearn.cluster import KMeans
4 from sklearn.metrics import silhouette_score
5
6 # JDS Clustering
7 X_j = StandardScaler().fit_transform(jds_core[JDS_SKILLS
8 ])

```

```

8 pca_j = PCA(n_components=3).fit(X_j)
9 sil_j = {k: silhouette_score(X_j, KMeans(k, random_state
10 =42).fit_predict(X_j)) for k in range(2, 7)}
11 jds_core['cluster'] = KMeans(3, random_state=42).
12 fit_predict(X_j)
13 prof_j = jds_core.groupby('cluster')[JDS_SKILLS + ['
14 outcome']].mean()
15
16 # SDS Clustering
17 X_s = StandardScaler().fit_transform(sds_core[SDS_TRAITS
18 ])
19 pca_s = PCA(n_components=3).fit(X_s)
20 sil_s = {k: silhouette_score(X_s, KMeans(k, random_state
21 =42).fit_predict(X_s)) for k in range(2, 7)}
22 sds_core['cluster'] = KMeans(4, random_state=42).
23 fit_predict(X_s)
24 prof_s = sds_core.groupby('cluster')[SDS_TRAITS + ['
25 success_classification_high_low']].mean()

```

Listing 20. PCA loadings, silhouette evaluation across k, and cluster centroid profiling.

```

JDS: explained variance
share
PC1 0.408
PC2 0.215
PC3 0.160
PC4 0.124
PC5 0.094JDS: loadings (PC1-PC3)
PC1 PC2 PC3
big_data -0.026 0.903 0.237
maths_stats 0.540 -0.206 -0.169
coding 0.509 0.193 -0.550
ai & ml 0.438 -0.207 0.779
dashboard & storytelling 0.506 0.251 0.073<Figure size 1200x450 with 2 Axes>JDS:
silhouette by k -> (2: np.float64(0.333), 3: np.float64(0.337), 4: np.
float64(0.283), 5: np.float64(0.263), 6: np.float64(0.315)) | chosen k=3
JDS: cluster profiles (pct_high = % in the high-outcome group)
big_data_skills maths_stats_skills coding_skills ai_and_ml_skills
dashboard_and_storytelling_skills n pct_high
cluster
0 3.813 3.639 4.348 3.970 3.230
1 3.831 4.721 4.815 4.827
2 3.911 4.877 83.333 3.326 4.839
3.687 38 18.421<Figure size 800x420 with 2 Axes>
SDS: explained variance
share
PC1 0.413
PC2 0.210
PC3 0.167
PC4 0.115
PC5 0.096SDS: loadings (PC1-PC3)
PC1 PC2 PC3
neuroticism -0.247 0.766 0.483
extraversion 0.533 -0.040 -0.227
openness_to_experience 0.471 0.513 -0.229
agreeableness 0.374 -0.341 0.810
conscientiousness 0.541 0.178 0.083<Figure size 1200x450 with 2 Axes>SDS:
silhouette by k -> (2: np.float64(0.313), 3: np.float64(0.322), 4: np.
float64(0.335), 5: np.float64(0.212), 6: np.float64(0.216)) | chosen k=4
SDS: cluster profiles (pct_high = % in the high-outcome group)
neuroticism extraversion openness_to_experience agreeableness
conscientiousness n pct_high
0 54.818 36.136 42.182 34.545
35.682 22 27.273
1 35.950 36.500 27.250 58.600
34.050 20 0.000
2 33.820 49.517 47.955 48.225
53.674 89 88.764
3 29.733 34.133 30.433 31.900
34.533 30 0.000<Figure size 800x460 with 2 Axes>

```

Listing 21. Execution Output G.10: PCA Loadings and Cluster Profiles.

G.11 Repeated Cross-Validation and Interpretable Classi- fiers: [← Return to SDS Models (V-A)] [← Return to JDS Models (V-B)]

```

1 from sklearn.model_selection import
2 RepeatedStratifiedKFold, cross_validate
3 from sklearn.linear_model import LogisticRegression
4 from sklearn.tree import DecisionTreeClassifier,
5 export_text
6 from sklearn.ensemble import RandomForestClassifier
7 from sklearn.inspection import permutation_importance
8
9 cv = RepeatedStratifiedKFold(n_splits=5, n_repeats=10,
10 random_state=42)

```

```

9 # Models evaluation function
10 def evaluate_cv(X, y):
11     models = {
12         'Baseline': DummyClassifier(strategy='
most_frequent'),
13         'Logistic': LogisticRegression(),
14         'Decision Tree': DecisionTreeClassifier(max_depth
=3),
15         'Random Forest': RandomForestClassifier(
n_estimators=300, random_state=42)
16     }
17     return {name: cross_validate(m, X, y, cv=cv, scoring
=['roc_auc', 'accuracy']) for name, m in models.
items()}

```

Listing 22. Repeated stratified 5-fold cross-validation and shallow decision tree rule extraction.

```

JDS: predict high salary hike: repeated 5-fold CV (10 repeats)
cv_auc auc_sd cv_accuracy
model
Baseline (majority) 0.500 0.000 0.525
Logistic (standardised) 0.903 0.053 0.857
Decision tree (depth 3) 0.800 0.082 0.782
Random forest 0.875 0.059 0.824JDS: logistic coefficients (per
1 SD)

std_coef odds_ratio_per_lsd
maths_stats_skills 1.284 3.612
dashboard_and_storytelling_skills 1.117 3.057
ai_and_ml_skills 0.762 2.143
big_data_skills 0.683 1.980
coding_skills 0.533 1.704JDS: random-forest
permutation importance (30% hold-out)

perm_importance sd
maths_stats_skills 0.091 0.038
dashboard_and_storytelling_skills 0.085 0.034
ai_and_ml_skills 0.030 0.026
big_data_skills 0.009 0.012
coding_skills 0.005 0.018JDS: depth-3 tree rules
|---- dashboard_and_storytelling_skills <= 4.15
| |---- maths_stats_skills <= 4.60
| | |---- class: 0
| | |---- maths_stats_skills > 4.60
| | |---- dashboard_and_storytelling_skills <= 2.55
| | | |---- class: 0
| | |---- dashboard_and_storytelling_skills > 2.55
| | | |---- class: 0
| |---- dashboard_and_storytelling_skills > 4.15
| |---- maths_stats_skills <= 3.55
| | |---- class: 0
| |---- maths_stats_skills > 3.55
| | |---- coding_skills <= 4.65
| | | |---- class: 1
| | |---- coding_skills > 4.65
| | | |---- class: 1

JDS sensitivity: logistic AUC all rows=0.903 | excluding flagged rows (n=113)=0.875

```

Listing 23. Execution Output G.11a: Junior Model CV and Decision Rules (Table VII).

```

SDS: predict high success: repeated 5-fold CV (10 repeats)
cv_auc auc_sd cv_accuracy
model
Baseline (majority) 0.500 0.000 0.528
Logistic (standardised) 0.964 0.033 0.927
Decision tree (depth 3) 0.942 0.039 0.934
Random forest 0.996 0.008 0.949SDS: logistic coefficients (per
1 SD)

std_coef odds_ratio_per_lsd
conscientiousness 2.094 8.114
openness_to_experience 2.043 7.717
extraversion 0.954 2.595
neuroticism 0.798 2.222
agreeableness 0.628 1.873SDS: random-forest permutation
importance (30% hold-out)

perm_importance sd
openness_to_experience 0.081 0.035
conscientiousness 0.044 0.021
agreeableness 0.021 0.009
extraversion 0.006 0.008
neuroticism 0.003 0.002SDS: depth-3 tree rules
|---- openness_to_experience <= 38.50
| |---- class: 0
|---- openness_to_experience > 38.50
| |---- conscientiousness <= 36.50
| | |---- class: 0
| |---- conscientiousness > 36.50
| | |---- agreeableness <= 37.50
| | | |---- class: 0
| | |---- agreeableness > 37.50
| | | |---- class: 1

SDS sensitivity: logistic AUC all rows=0.964 | excluding flagged rows (n=143)=0.955

```

Listing 24. Execution Output G.11b: Senior Model CV and Decision Rules (Table V).

G.12 Machine Learning Salary Regression Baselines: [[← Return to Salary Regressions in Section V-F](#)]

```

1 from sklearn.linear_model import Ridge
2 from sklearn.ensemble import
HistGradientBoostingRegressor
3 from sklearn.model_selection import KFold
4
5 kf = KFold(n_splits=5, shuffle=True, random_state=42)
6
7 # DataScience Jobs Model
8 X_dsj = pd.get_dummies(dsj_core[['min_experience', '
role_family', 'num_of_jobs', 'is_senior_title']],
drop_first=True)
9 y_dsj = np.log(dsj_core['avg_salary_lpa'])
10 ridge_dsj = cross_validate(Ridge(), X_dsj, y_dsj, cv=kf,
scoring=['r2', 'neg_mean_absolute_error'])
11
12 # Analytics Jobs Model
13 X_an = pd.get_dummies(an_core[['exp_min_yrs', '
role_family', 'location_group']] + [c for c in
an_core if c.startswith('skill_')]), drop_first=True)
14 y_an = an_core['salary_mid']
15 gbr_an = cross_validate(HistGradientBoostingRegressor(
random_state=42), X_an, y_an, cv=kf, scoring=['r2',
'neg_mean_absolute_error'])

```

Listing 25. Cross-validated Ridge and HistGradientBoosting regressions with permutation importances.

```

DataScience Jobs salary model (target=log avg_salary_lpa)
cv_r2 cv_mae
model
Baseline (mean) -0.002 0.487
Ridge 0.571 0.305
Gradient boosting 0.552 0.311
permutation_importance (R2 drop)
min_experience 0.504
role_family 0.435
log_jobs 0.044
is_senior_title 0.018
large_recruiter 0.004Analytics Jobs salary model (
target=salary_mid)
cv_r2 cv_mae
model
Baseline (mean) -0.001 6.972
Ridge 0.439 5.022
Gradient boosting 0.444 4.989
permutation_importance (R2 drop)
exp_min_yrs 0.798
role_family 0.021
location_group 0.014
skill_ai_and_ml_skills 0.004
n_skill_dims 0.004
skill_maths_stats_skills 0.003
is_metro 0.001
skill_coding_skills 0.000
skill_dashboard_and_storytelling_skills 0.000
skill_big_data_skills -0.000Read these
as descriptive drivers of advertised salary, not causal effects.

```

Listing 26. Execution Output G.12: Salary Regression Benchmarks and Feature Importance.

G.13 Export of Reproducible Artifacts: [[← Return to Reproducibility in Section IV](#)]

```

1 exports = {
2     'jds_core': jds_core, 'sds_core': sds_core,
3     'dsj_core': dsj_core, 'analytics_core_no_text':
an_core,
4     'table_jds_effects': jds_effects, 'table_sds_effects'
: sds_effects,
5     'table_dsjsenior_premium': prem, '
bridge_skill_alignment': alignment
6 }
7 for name, df in exports.items():
8     df.to_csv(f'{name}.csv', index=True)
9 shutil.make_archive('dd_foundation_outputs', 'zip',
OUT_DIR)
10 print('Saved all artifacts successfully.')

```

Listing 27. Automated serialization of clean tables, bridge matrices, and figures.

Saved 12 tables + 17 figures to outputs -> dd_foundation_outputs.zip

Listing 28. Execution Output G.13: Export Archive Generation Trace.

G.14 Final Analytical Recap and Verification: [[← Return to Analytical Results in Section VI](#)]

```
1 print('=' * 78)
2 print('FOUNDATION RECAP')
3 print('=' * 78)
4 print(f'Core rows -> JDS {len(jds_core)}, SDS {len(
    sds_core)}, DS Jobs {len(dsj_core)}, Analytics {len(
    an_core):,}')
5 print(f'Strongest junior skill signal : {jds_effects.loc
    [0, "feature"]} (d={jds_effects.loc[0, "cohens_d
    "]:.2f})')
6 print(f'Strongest senior trait signal : {sds_effects.loc
    [0, "feature"]} (d={sds_effects.loc[0, "cohens_d
    "]:.2f})')
7 print(f'Median senior salary premium across role families
    : {prem["senior_premium_pct"].median():.0f}%')
8 print(f'Largest importance-vs-demand gap: {alignment.
    sort_values("rank_gap", ascending=False).iloc[0]["
    feature"]}')

```

Listing 29. One-screen summary log confirming statistical signal integrity.

```
=====
FOUNDATION RECAP
=====
Core rows -> JDS 139, SDS 161, DS Jobs 1595, Analytics 14,749
Strongest junior skill signal : dashboard & storytelling (d=1.30)
Strongest senior trait signal : conscientiousness (d=1.81); weakest: neuroticism (d
=-0.01)
JDS model CV AUC (logistic) 0.90 | SDS 0.96
Median senior salary premium across role families: 64%
Largest importance-vs-demand gap: dashboard & storytelling (rank gap +2)

```

Listing 30. Execution Output G.14: Final Terminal Recap Log.

F. Acknowledgments

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